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ORIGINAL RESEARCH

Advancing firefighter games: novel integer programming formulations and the cost-value model

Marta Baldomero-Naranjo[1] · **Jörg Kalcsics**[2] · **Antonio M. Rodríguez-Chía**[1] · **Catriona Wedderburn**[3]

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Abstract
In the classical firefighter game, a fire breaks out on some vertices of an undirected connected graph at time zero. At each subsequent time step, a fixed number of firefighters can protect one vertex each from catching fire. Afterwards, the fire spreads from each burning vertex to every adjacent vertex that is neither burning nor defended. The game ends when the fire can no longer spread. The goal is to find a defense strategy that maximizes the number of non-burning (saved) vertices. In this work, we first revisit the classical integer linear programming formulation and then present several improvements for it, as well as two new formulations and tighter bounds on the maximum duration of the game. Moreover, we relax the classical assumptions that all vertices have uniform values and costs, i.e., we allow vertices to have different values and costs for being defended. Furthermore, instead of a fixed number of firefighters, we are given a defense budget that we can spend each time step to defend the vertices. We call this the cost-value firefighter game. We present three different integer linear programming formulations for the problem, along with a series of inequalities to strengthen the formulations and tight bounds on the maximum duration of the game.

Keywords Firefighter game · Mixed integer programming · Combinatorial optimization

✉ Jörg Kalcsics
joerg.kalcsics@ed.ac.uk

Marta Baldomero-Naranjo
marta.baldomero@uca.es

Antonio M. Rodríguez-Chía
antonio.rodriguezchia@uca.es

Catriona Wedderburn
cat.wedderburn@gmail.com

[1] Departamento de Estadística e Investigación Operativa, Universidad de Cádiz, Facultad de Ciencias, 11510 Puerto Real (Cádiz), Spain

[2] School of Mathematics and the Maxwell Institute for Mathematical Sciences, University of Edinburgh, EH9 3FD Edinburgh, United Kingdom

[3] Royal (Dick) School of Veterinary Studies, Roslin Institute, and Global Academy of Agriculture and Food Systems, University of Edinburgh, Edinburgh EH25 9RG, United Kingdom

1 Introduction

The Classical Firefighter Game (CFG) is a discrete-time, deterministic model that abstracts the dynamics of fire spreading on a network, where strategic protection of vertices is used to contain the fire, first introduced by Hartnell (1995). In this problem, a 'fire' breaks out in a fixed set of vertices of a connected undirected graph at time step 0. A vertex catching fire is called 'burning'. In each subsequent time step, first, a certain number of vertices can be defended before the fire spreads to all non-burning undefended neighbors of the burning vertices. Once a vertex is either burning or defended, it remains so for the rest of the game. The game ends when the fire can not spread further. The goal of the defenders is to find a strategy that maximizes the number of non-burning vertices. The decision version of CFG with one firefighter and starting fire was shown in MacGillivray and Wang (2003) to be NP-complete even with restricted to bipartite graphs and in Finbow et al. (2007) to be NP-complete even on trees with a maximum degree of three if the fire breaks out on a vertex of degree three.

CFG can also be seen as an abstraction of various disease intervention strategies, including administering a vaccine that prevents the vertex from being infected or from spreading the disease to other vertices; or an intervention that simply removes individuals from the area and thus from the path of the disease. Such interventions have been used, for example, to save endangered species from the chrytridiomycosis pandemic, see (Goodman, 2006). The vertices hereby model individuals, with edges connecting individuals between whom disease could spread (for instance, if they regularly come within a certain physical distance for airborne transmission). Alternatively, the vertices could model population centers, with edges joining centers that in real life are linked by transportation and regularly have individuals traveling between them, potentially spreading the disease. Although contact tracing and ring vaccination (vaccinating everyone who came into contact with an infected individual and everyone who came into contact with them etc.) within population centers are frequently used, sometimes it is preferable to vaccinate entire populations. This could be due to a lack of information about contacts; a policy choice based on the cost of vaccines and the risk to the public; or the logistical difficulty of having to visit the same population centers multiple times to administer additional vaccines. Other applications include, e.g., stopping a virus from spreading on a network (Finbow et al., 2000) or hampering the spread of fake news on social media networks. Furthermore, operations research plays a significant role in addressing open questions in fire management techniques, as stated in Rönnqvist et al. (2015), Rönnqvist et al. (2023). In fact, it has been widely applied to optimize resource allocation in firefighting strategies, as seen, for example, in Mendes and e Alvelos (2025), Ozkan and Kilic (2023), Yang et al. (2019).

As our first contribution, we revisit the classical Integer Linear Programming (IP) formulation for CFG and introduce several improvements, including inequalities to strengthen the formulation. Moreover, we derive tighter bounds on the duration of the game that significantly reduce the size of the formulation. Finally, we propose two new IP formulations for CFG that empirically turn out to be much more efficient than the classical formulation.

As a second contribution, we discuss a variant of this problem: the Cost-Value Firefighter game (CVFG). In CVFG, every vertex has both a cost of defense and a utility gained by protecting that vertex from catching fire. Moreover, instead of a fixed number of firefighters, a defense budget is given that can be spent each time step for defending vertices. The cost for defending a vertex has to be paid only once, and the vertex remains defended for the rest of the game. In addition, and differing from the classic game, in this variant the defense of a vertex can be built up gradually, i.e., over multiple time steps. However, a vertex is only safe

from catching fire when full protection has been achieved. The objective is then to maximize the total utility of the non-burning vertices. CVFG can better represent some real-word applications by setting: i) the cost is proportional to the effort needed to defend a given area from catching fire—or, equivalently, to the population size that must be vaccinated— and any extra costs related to that location; and ii) the value proportional to the the size/importance of the area that would be protected. There has been some recent interest in modeling the cost of vaccinating different vertices, notably in Michalak (2020), whose model is based on CFG but uses probabilistic disease spread and control interventions beyond vaccination. Moreover, CVFG allows us to easily model the case where the fire takes more than one period to spread between two adjacent nodes. To do this, we simply insert as many auxiliary nodes into the edge as the time it takes for the contagion to spread along the edge minus one. For such auxiliary nodes, we set the value of saving the node equal to zero and its defense cost to a very large constant. We present a first integer programming formulation for CVFG that is derived from the classical formulation for CFG. Afterwards, we introduce several improvements to strengthen the formulation, tight bounds on the maximum duration of the game, and two new integer programming formulations. We present an extensive computational study to compare the formulations and assess the suggested improvements.

The remainder of the paper is organized as follows. In Section 2, we review the related literature. In the next section, we first present the classical integer programming formulation for the firefighter problem. Afterwards, we introduce several improvements to strengthen the formulation and two new integer programming formulations. We also present extensive computational results to analyze the efficiency of the formulations and the proposed improvements. Section 4 mirrors the previous section, but now for CVFG. Finally, our conclusions and some future research topics are included in Section 5.

2 Literature review

In the following, we provide an overview of the related literature on the firefighter problem. Concerning methodological contributions, we will focus on exact algorithms, which are the scope of this paper, and we will only briefly mention recent heuristic algorithms.

The first IP formulation for CFG was introduced in Develin and Hartke (2007) and modified in Finbow and MacGillivray (2009) by adding one set of valid inequalities. To derive the formulation, an upper bound on the maximum duration of the game is needed. In García-Martínez et al. (2015), the authors carry out an empirical study to assess the impact of the value of the upper bound on the run times of the IP formulation using the formulation proposed in Develin and Hartke (2007). Not unsurprisingly, they observed that the closer the upper bound is to the actual duration of the game, the faster the IP formulation can be solved, stressing for the first time the importance of tight upper bounds. Moreover, they also proposed a parametric IP strategy that automatically adjusts the value of the upper bound. However, no formal optimality criterion is provided for the value of the upper bound, rendering it a heuristic. The authors also propose several other heuristics for the problem. Two improvements to the IP formulation of Develin and Hartke (2007) are presented in Ramos et al. (2020). Firstly, they propose a preprocessing step to fix variables and use an aggregate version of one of the constraints. Secondly, they derive a slightly improved upper bound. In addition, they propose a matheuristic to solve the problem. Other heuristics for CFG can be found, for example, in Blum et al. (2014) and Hu et al. (2015). We are not aware of any

further work on strengthening the IP formulation of Develin and Hartke (2007) or providing alternative formulations.

All these works are applicable to general networks. For specialized graphs, often better algorithms or results can be obtained. For example, for graphs with bounded tree width (Cai et al., 2010; Enright & Meeks, 2015), for planar graphs (Esperet et al., 2013; Gordinowicz, 2015; Wang et al., 2014), for trees (Coupechoux et al., 2019; Iwaikawa et al., 2011), for different kinds of grids (Chen et al., 2017; Develin & Hartke, 2007; Gavenčiak et al., 2014), or for graphs with certain node degrees (Finbow et al., 2007; Prałat, 2014). Moreover, most works deal with static networks, i.e., graphs whose structure does not change, and spread is assumed to be deterministic, i.e., every non-burning vertex adjacent to a burning vertex will catch fire in the next round. Concerning the former assumption, in recent years, there has been research on dynamic networks whose structure can change over time, see, e.g., (Enright & Kao, 2018; Pellis et al., 2015). And with respect to the latter, in Michalak (2020), the authors consider the situation in which a disease (i.e., fire) spreads probabilistically across the graph and the goal is to minimize the combined cost of infections, vaccinations, and (vertex) isolation. The authors propose an evolutionary algorithm to solve the problem. Other probabilistic approaches can be found, e.g., in Tennenholtz et al. (2017). For a more comprehensive review of the literature on the firefighter problem, see (Finbow & MacGillivray, 2009; Wagner, 2021).

Concerning the cost-value firefighter game, in Duffy and MacGillivray (2015), the authors introduce the weighted firefighter problem, which associates a weight with each vertex, and the goal is to maximize the total weight of all non-burning vertices at the end of the game. The authors discuss this problem for one starting fire and one firefighter and prove that it is NP-hard even on binary trees. A version where vertices can have arbitrary positive or negative weights and the goal is to maximize the sum of weights of non-burning vertices minus the sum of weights of burning vertices is considered in Costa et al. (2016). They prove that the problem is NP-hard on binary trees and derive two greedy approximation algorithms on trees. In a slightly different vein, in Coupechoux et al. (2019), the authors introduce the fractional firefighter game. While all vertices have a cost and value of one, it is allowed to split the defense budget to only partially, i.e., fractionally, defend a vertex in a given time step. A partially defended vertex can still catch fire, but only the unprotected portion will count as burning. Moreover, a partially burning vertex will only spread this portion of the fire to its neighbors. The problem formulation was motivated by an online setting, where the defense budget can vary, and one does not know how much will be available in future time steps. We are not aware of any IP formulation or computational study for CVFG.

3 Classic firefighter

We start by recounting the classic firefighter game (CFG) and its IP formulation. Let $G = (V, E)$ be a simple undirected graph with vertex set V and edge set $E \subseteq V \times V$. Moreover, let $N(i) = \{j \in V : [i, j] \in E\}$ be the set of vertices adjacent to $i \in V$ and let $dist(i, j) \in \mathbb{Z}^+$ denote the shortest path distance between $i, j \in V$ (each edge has length one). In addition, for $W \subseteq V$ let $N(W) = \cup_{i \in W} N(i)$ and $dist(i, W) = \min_{j \in W} dist(i, j)$. Finally, let $F \subset V$ and $f = |F|$ be the set and number of initially burning vertices, respectively, and $d \in \mathbb{Z}^+$ be the number of available firefighters. The game adheres to the following four rules:

1. All vertices in F are *burning* at time 0.
2. In each subsequent time step $t > 0$, we first *defend* up to d previously undefended vertices and afterwards the fire spreads from every burning vertex to all adjacent *susceptible* vertices, i.e., vertices that are neither burning nor defended.
3. Once burning (defended), a vertex will remain burning (defended) until the end of the game.
4. The game ends in time step $T \in \mathbb{Z}^+$ if either
   - a node ignites at time T, but the fire cannot spread any further in the following time steps independent of what happens defense-wise, i.e., even if we stop defending vertices all together, or
   - no node ignites at time T, but a node ignited at time $T-1$ and without defending in time T, the fire would have continued to spread.

The goal of the game is to maximize the number of non-burning vertices at the end of the game; or equivalently, minimize the number of burning vertices. An (optimal) solution to the game is called an *(optimal) defense strategy*. For an optimal defense strategy, we denote by T^{opt} the time step when the game ends. If there is more than one optimal defense strategy and they have different values for T^{opt}, then we define T^{opt} as the smallest among all values, i.e., we choose a defense strategy that ends the game the quickest. Unfortunately, even for T^{opt} there are potentially many alternative optimal solutions. However, we can always find one that satisfies the following:

Lemma 3.1 *For any instance of CFG, there exists an optimal defense strategy S that uses every firefighter at time step $t \leq T^{opt} - 1$ and only defends vertices that must be defended (i.e., which would otherwise burn before the game ends).*

Proof Let S be an optimal defense strategy for an instance of CFG that ends the game after T^{opt} time steps. Assume that S does not use all firefighters at time $t \leq T^{opt} - 1$ or firefighters defend vertices that would not have needed to be defended. As the game does not end in t and T^{opt} is the minimum, we need to distinguish two cases:

Case 1: There is at least one susceptible vertex $j \in V$ at time T^{opt} that still needs to be defended, but no new vertices catch fire. Let $W \subseteq V$ denote the set of vertices that need to be defended in T^{opt}, i.e., at the end of the game.
If S uses fewer than d firefighters at some time step $t \leq T^{opt} - 1$, then the unused firefighters could already defend the vertices in W, resulting in an alternative defense strategy S' which has the same burning and also ends after T^{opt} steps. Note that we must have strictly less than $|W|$ unused firefighters in total up to and including period $T^{opt} - 1$, because otherwise we could have completely defended all nodes in W before T^{opt} and the game would already have ended at $T^{opt} - 1$. Hence, all unused firefighters before time T^{opt} can be assigned to defend vertices in W. On the other hand, if S defends one or more vertices at time t that would not need to be defended, then again we could use those firefighters to defend vertices in W instead. Again, we must have less than $|W|$ firefighters in total defending such vertices, and all of them can be reassigned to vertices in W.

Case 2: At least one new vertex $j \in V$ is igniting at time T^{opt}, but then the fire is fully contained without further defending. Let $W \subseteq V$ denote the set of vertices catching fire in T^{opt}. As S is an optimal strategy, we cannot have any unused firefighters in periods $t \leq T^{opt} - 1$ or firefighters defending vertices that did not need to be defended. Because, otherwise, we could re-assign those firefighters to defend the vertices in W, resulting in a solution that has strictly less burning than S. □

As a consequence of Lemma 3.1, we can assume for the remainder that any optimal defense strategy uses all firefighters and only assigns them to necessary-to-defend vertices in every time step $t < T^{opt}$.

3.1 Integer programming formulation

In the following, we first present the integer linear programming formulation for CFG as introduced in Develin and Hartke (2007). Let $T \in \mathbb{Z}^+$ be an upper bound for T^{opt} (a trivial one is $T = n$; we will propose much better values for T in Section 3.3). The authors defined the decision variables

$$b_i^t = \begin{cases} 1, & \text{if vertex } i \text{ is burning at time } t, \\ 0, & \text{otherwise,} \end{cases}$$

$$d_i^t = \begin{cases} 1, & \text{if vertex } i \text{ is defended at time } t, \\ 0, & \text{otherwise,} \end{cases}$$

for all $i \in V$ and $0 \le t \le T$. In the following, we call b_i^t and d_i^t the *fire* and *defense variables*, respectively. With these variables, they formulated the problem as

$$\text{(CF)} \qquad \min \sum_{i \in V} b_i^T \tag{3.1a}$$

$$\text{s.t.} \quad b_i^{t-1} \le b_i^t, \qquad i \in V,\ 1 \le t \le T, \tag{3.1b}$$

$$d_i^{t-1} \le d_i^t, \qquad i \in V,\ 1 \le t \le T, \tag{3.1c}$$

$$b_i^{t-1} \le b_j^t + d_j^t, \qquad i \in V,\ j \in N(i),\ 1 \le t \le T, \tag{3.1d}$$

$$b_i^t + d_i^t \le 1, \qquad i \in V,\ 1 \le t \le T, \tag{3.1e}$$

$$\sum_{i \in V} \left(d_i^t - d_i^{t-1} \right) \le d, \qquad 1 \le t \le T, \tag{3.1f}$$

$$b_i^0 = 1, \qquad i \in F, \tag{3.1g}$$

$$d_i^0 = 0, \qquad i \in V, \tag{3.1h}$$

$$b_i^t,\ d_i^t \in \{0, 1\}, \qquad i \in V,\ 0 \le t \le T. \tag{3.1i}$$

The objective function (3.1a) minimizes the number of burning vertices. The permanency of burning and defense is ensured by Constraints (3.1b) and (3.1c), respectively. A vertex j neighboring a burning vertex i must either burn or be defended in the following time step due to Constraints (3.1d). That a vertex i cannot burn and be defended at the same time is asserted by Constraints (3.1e). Constraints (3.1f) ensure that no more than d vertices are defended each time step. Fires start in the vertices of F at time 0 by Constraints (3.1g) and no vertices are initially defended by Constraints (3.1h). Finally, burning and defense variables are binary by Constraints (3.1i).

3.2 Improvements to the formulation

In this section, we review existing formulations and present new improvements for CF, which will then be thoroughly tested in Section 3.5.

3.2.1 Preprocessing

As suggested in Ramos et al. (2020), we can fix all fire variables b_i^t to 0 for $t < \min_{j \in F} dist(i, j)$, since a fire cannot possibly reach node i by time t, if i is more than t away from any fire in F.

3.2.2 Changes to the constraint sets

As a second contribution, in Ramos et al. (2020), the authors suggest aggregating the defense constraints (3.1f) over time, bounding the total number of defended vertices so far by

$$\sum_{i \in V} d_i^t \leq t \cdot d, \quad 1 \leq t \leq T, \tag{3.2}$$

which also allows dropping (3.1h). The authors prove that an optimal solution of formulation (3.1a) − (3.1e), (3.1g), (3.1i), and (3.2) can be converted into an optimal solution for formulation (3.1a) − (3.1i).

To ensure that vertex i can only be burning at time t if one of its neighbours was already burning in the previous period, i.e., vertices cannot spontaneously ignite, in Finbow and MacGillivray (2009), the authors added

$$b_i^t \leq \sum_{j \in N(i)} b_j^{t-1} \quad i \in V,\ 1 \leq t \leq T. \tag{3.3}$$

Next, we suggest some additional improvements. We observe that Constraints (3.1e) are not strictly necessary and can therefore be dropped. If $d_i^t = 1$, then setting $b_i^t = 0$ is feasible on the right-hand side of (3.1d) and the objective function will do so whenever possible. And if $b_i^t = 1$, then there is no point in defending i as the vertex is already burning. Instead, the firefighter should be used on a different vertex or not at all, if there are no more susceptible vertices, see Lemma 3.1. Moreover, due to (3.1b) and (3.1c), it would be sufficient to add them just for the final period

$$b_i^T + d_i^T \leq 1, \qquad i \in V. \tag{3.4}$$

Finally, a compromise between the two is adding them only for the first $\lceil \alpha \cdot T \rceil$ periods, $\alpha \in [0, 1]$. We denote this version of the constraints as "α(3.1e)".

3.2.3 Changes to variable domains

Fire and defense variables are defined as binary in CF. However, we can relax integrality on either set of variables without invalidating the correctness of the formulation.

Lemma 3.2 *If we remove the binary constraints on the fire variables in CF while keeping the defense variables binary, then we can convert any optimal solution to the relaxed formulation into one in which all fire variables have binary values. Analogously, if we instead relax the binary constraints on the defense variables while keeping the fire variables binary, the same property holds: any optimal solution can be converted into one where all defense variables are binary.*

Proof We first relax the fire variables. Let $(\hat{b}, \hat{d})$ be an optimal solution of this relaxed formulation and assume that at least one fire variable is fractional. Let $1 \leq t \leq T$ be the smallest time step such that there exists a $j \in V$ with $0 < \hat{b}_j^t < 1$. As t is the minimum, we

must have $\hat{b}_i^{t-1} \in \{0, 1\}$ for all $i \in N(j)$. On the one hand, if none of the adjacent vertices $i \in N(j)$ are burning at $t-1$, i.e., $\hat{b}_i^{t-1} = 0$, then we can set $\hat{b}_j^t = 0$ without violating (3.1d) or losing optimality. On the other hand, if at least one $\hat{b}_i^{t-1} = 1$, then (3.1d) implies that $\hat{d}_i^t = 1$ as defense variables are still binary. Hence, setting $\hat{b}_j^t = 0$ would yield a solution that is at least as good.

Next, we relax the defense variables. Let $(\hat{b}, \hat{d})$ be an optimal solution of this relaxed formulation and assume that at least one defense variable is fractional. Let $1 \leq t \leq T$ be the last time step such that there exists a vertex $j \in V$ with $0 < \hat{d}_j^t < 1$. In the following, we show how to reduce the number of fractional defense variables in t by at least one. If we have some leftover amount of firefighters in t, i.e., $\Delta^t = d - \sum_{i \in V} \hat{d}_i^t > 0$, then we can increase $\hat{d}_j^t$ by $\min\{\Delta^t, 1 - \hat{d}_j^t\}$ ("amount", because Δ^t can be fractional). If we get $\hat{d}_j^t = 1$ by doing so, then we have found an alternative optimal solution but with one less fractional defense variable at time t. Otherwise, $\hat{d}_j^t$ remains fractional, but now with $\Delta^t = 0$. Concerning the latter case, on the one hand, if at least one adjacent vertex $i \in N(j)$ is burning at $t-1$, i.e., $\hat{b}_i^{t-1} = 1$, then we must have $\hat{b}_j^t = 1$. Thus, there is no point in defending j and we can set $\hat{d}_j^r = 0$, $r = 1, \ldots, T$, without violating (3.1d) or losing optimality. On the other hand, let $\hat{b}_i^{t-1} = 0$ for all $i \in N(j)$. If $t = T$, then we can simply set $\hat{d}_j^t = 0$. Otherwise, since t is maximal, we have $\hat{d}_j^{t+1} = 1$ by constraints (3.1c), i.e., j will be fully defended in $t+1$. Let $\Delta^{t+1} = d - \sum_{i \in V} \hat{d}_i^{t+1} \geq 0$ be the amount of leftover firefighters in $t+1$. If $\hat{d}_j^t \leq \Delta^{t+1}$, then we have enough firefighters left to fully defend j from scratch at time $t+1$, i.e., we can set $\hat{d}_j^{t'} = 0$, $t' \leq t$, and we have one less fractional variable at time t. Thus, let now $\Delta^{t+1} < \hat{d}_j^t$. As d is integer and t is maximum, there must be at least one other vertex j' with $\hat{d}_{j'}^{t+1} = 1$ and a fractional increase of defense at time $t+1$, i.e., $0 < \delta_{j'} = \hat{d}_{j'}^{t+1} - \hat{d}_{j'}^t < 1$. Moreover, we must have $0 < \hat{d}_{j'}^t < 1$. Now, as j cannot catch fire in t, we can transfer the fraction of defense $\hat{d}_j^t$ spent on j before $t+1$ to j' until either $\hat{d}_j^t = 0$ or $\hat{d}_{j'}^t = 1$. In either case, we have found an alternative optimal solution, but with one less fractional defense variable at time t. Repeating the argument completes the proof. □

3.3 Bounds on the maximum duration of the game

Concerning the upper bound T, an obvious improvement over $T_0^{max} = n$ is given by $T_1^{max} = n - f$, the total number of initially non-burning vertices. In Ramos et al. (2020), the authors propose to use $T_2^{max} = \lceil \frac{n}{d} \rceil$, i.e., the maximum number of time steps needed to defend every single vertex (the authors propose an even tighter bound, however, without giving a proof that the bound is valid, thereby sacrificing the correctness of the formulation and rendering it a matheuristic). A trivial improvement combines the above two: $T_3^{max} = \lceil \frac{n-f}{d} \rceil$. An even tighter bound can be derived from the observation that, for the game to continue, at least one new vertex must catch fire every time step. Hence, each time step at least $d+1$ vertices are newly defended or burning, so the game must end after at most $T_4^{max} = \lceil \frac{n-f}{d+1} \rceil$ time steps. Observe that the computed bound replaces the parameter T in the formulation.

3.4 New formulations for the classic firefighter game

Next, we present our two new formulations.

3.4.1 The indicator formulation

The idea of the first is to replace the status variables b_i^t and d_i^t by variables that indicate instead the time period when the status of a node changes

$$\hat{b}_i^t = \begin{cases} 1, & \text{if vertex } i \text{ is catching fire for the first time at time } t, \\ 0, & \text{otherwise,} \end{cases}$$

$$\hat{d}_i^t = \begin{cases} 1, & \text{if vertex } i \text{ is being defended for the first time at time } t, \\ 0, & \text{otherwise,} \end{cases}$$

for all $i \in V$ and $0 \leq t \leq T$ for $\hat{b}_i^t$ and $1 \leq t \leq T$ for $\hat{d}_i^t$. With these variables, we can formulate CFG as

$$(\mathrm{CF}_1) \quad \min \quad \sum_{i \in V} \sum_{t=0}^{T} \hat{b}_i^t \tag{3.5a}$$

$$\text{s.t.} \quad \sum_{i \in V} \hat{d}_i^t \leq d, \qquad 1 \leq t \leq T, \tag{3.5b}$$

$$\sum_{r=0}^{t-1} \hat{b}_i^r \leq \sum_{r=0}^{t} \hat{b}_j^r + \sum_{r=1}^{t} \hat{d}_j^r, \qquad i \in V,\ j \in N(i),\ 1 \leq t \leq T, \tag{3.5c}$$

$$\hat{b}_i^0 = 1, \qquad i \in F, \tag{3.5d}$$

$$\hat{b}_i^t \in \{0, 1\}, \qquad i \in V,\ 0 \leq t \leq T, \tag{3.5e}$$

$$\hat{d}_i^t \in \{0, 1\}, \qquad i \in V,\ 1 \leq t \leq T. \tag{3.5f}$$

The objective function (3.5a) minimizes the number of burning vertices. Constraints (3.5b) ensure that no more than d vertices are defended in each time step. A vertex j adjacent to a burning vertex i must be burning or defended the time step after its neighbor first burned due to Constraints (3.5c). In order to tighten the constraint, we sum the fire variables over all $r \leq t - 1$ on the left side. Finally, fires start on the vertices of F at time 0 by (3.5d) and burning and defense are binary by (3.5e) and (3.5f).

Improvements to the formulation

Similar to CF, we can fix the fire variables $\hat{b}_i^t$ to 0 if $t < dist(i, F)$ and replace (3.5b) by the aggregated constraints

$$\sum_{i \in V} \sum_{r=1}^{t} \hat{d}_i^r \leq t \cdot d, \qquad 1 \leq t \leq T. \tag{3.6}$$

Next, we will introduce a series of inequalities to strengthen the formulation. Often, they are not valid in the strict sense, as they cut off feasible integer solutions. However, for every solution, these inequalities cut off, we show that there exists another feasible integer solution that is at least as good and is not cut off. While it is often intuitively clear that this holds, the formal proofs can be tedious and disrupt the flow of the paper. In such cases, we include the proof in the Appendix. For ease of writing, we will denote the solutions that are cut off by these inequalities as *dominated* in the following.

Our first sets of inequalities to strengthen CF_1 are

$$\sum_{r=0}^{t} \hat{b}_i^r \leq 1, \quad i \in V,\ 0 \leq t \leq T, \quad \text{and} \quad \sum_{r=1}^{t} \hat{d}_i^r \leq 1, \quad i \in V,\ 1 \leq t \leq T, \tag{3.7}$$

which limit the number of times a node can be defended or catch fire for the first time. Both inequalities only cut off dominated solutions. To see that, assume that t is the smallest time period such that $\sum_{r=0}^{t} \hat{b}_j^r = 2$ for some $j \in V$, i.e., $\hat{b}_j^{t'} = \hat{b}_j^t = 1$, $t' < t$. But then, as t is the minimum, the left-hand side of (3.5c) for every $i \in N(j)$ and t must be smaller than or equal to 1. Thus, there is no need to set $\hat{b}_j^t = 1$, because $\sum_{r=0}^{t-1} \hat{b}_j^r = 1$ and this only increases the objective function value. A similar argument applies for the second inequality. Moreover, adapting (3.1e), we can derive

$$\sum_{r=0}^{t} \hat{b}_i^r + \sum_{r=1}^{t} \hat{d}_i^r \leq 1, \quad i \in V,\ 1 \leq t \leq T, \tag{3.8}$$

which rule out that a node is defended and burning at the same time. We note that this inequality combines both inequalities in (3.7), and again only cuts off dominated solutions (follows from the discussions about (3.8) and relaxing (3.1e)). As for CF, in order to reduce the number of these inequalities, we can instead add them just for the final time step

$$\sum_{t=0}^{T} \hat{b}_i^t \leq 1 \quad \text{and} \quad \sum_{t=1}^{T} \hat{d}_i^t \leq 1, \qquad i \in V, \tag{3.9}$$

and

$$\sum_{t=0}^{T} \hat{b}_i^t + \sum_{t=1}^{T} \hat{d}_i^t \leq 1, \qquad i \in V, \tag{3.10}$$

respectively, or just for the first $\lceil \alpha\, T \rceil$ periods, which we will denote by "α(3.7)" and "α(3.8)", respectively.

Finally, we can again relax the integrality on either the defense or fire variables (the proof follows analogously to Lemma 3.2 using the equivalence $b_i^t = \sum_{r=0}^{t} \hat{b}_i^t$ and $d_i^t = \sum_{r=1}^{t} \hat{d}_i^t$).

3.4.2 The fire-path formulation

Our second formulation follows a similar idea of modeling the moment of change; however, this time we capture the path that the fire takes to spread. As it does not matter whether a node catches fire from only one neighbor or from multiple neighbors, we assume without loss of generality that a fire reaches a node from exactly one neighbor. The motivation for this is that for any optimal solution, the set of burning vertices and the edges along which the fire spreads will form a forest, with each tree of the forest rooted at an initially burning vertex. The variables modelling the fire spread are defined as

$$x_{ij}^t = \begin{cases} 1, & \text{if the fire spreads from vertex } i \text{ to vertex } j \text{ at time } t, \\ 0, & \text{otherwise,} \end{cases}$$

for all $i \in V$, $j \in N(i)$, and $1 \leq t \leq T$. Apart from that, we reuse the defense variables from the initial formulation.

$$(\text{CF}_2) \quad \min \sum_{i \in V} \sum_{j \in N(i)} \sum_{t=1}^{T} x_{ij}^t \tag{3.11a}$$

s.t. (3.1c),(3.1f), (3.1h)

$$1 \leq d_i^1 + \sum_{k \in N(i) \cap F} x_{ki}^1, \qquad i \in N(F) \setminus F, \tag{3.11b}$$

$$\sum_{k \in N(i), k \neq j} x_{ki}^{t-1} \leq x_{ij}^t + d_j^t + \sum_{r=\max\{1,t-2\}}^{t} \sum_{k \in N(j) \setminus \{i\}} x_{kj}^r, \qquad i \in V \setminus F,\ j \in N(i),\ 2 \leq t \leq T, \tag{3.11c}$$

$$x_{ij}^1 = 0, \qquad i \in V \setminus F,\ j \in N(i), \tag{3.11d}$$

$$x_{ij}^t \leq \sum_{k \in N(i) \setminus \{j\}} x_{ki}^{t-1}, \qquad i \in V \setminus F,\ j \in N(i),\ 2 \leq t \leq T, \tag{3.11e}$$

$$d_i^t \in \{0, 1\}, \qquad i \in V,\ 0 \leq t \leq T, \tag{3.11f}$$

$$x_{ij}^t \in \{0, 1\}, \qquad i \in V,\ j \in N(i),\ 1 \leq t \leq T. \tag{3.11g}$$

The objective function (3.11a) minimizes the number of burning vertices by counting how many vertices the fire spreads to. Constraints (3.1c), (3.1f), (3.1h) are taken from CF and model the correct behavior of firefighters. At time $t = 1$, a vertex i adjacent to an initially burning vertex must be defended at $t = 1$ or the fire spreads to i along one of its incident edges by Constraints (3.11b). We need to sum over all $k \in N(i) \cap F$ on the right-hand side as i could be adjacent to more than one vertex in F (otherwise, the final solution will no longer resemble a forest). For $t \geq 2$ and two adjacent vertices $i, j \in V$ with $i \notin F$, if the fire reaches vertex i at time $t - 1$ (left-hand side), then at time t the fire spreads along the edge to j (first term on the right) unless j is either defended (second term) or has caught fire from somewhere else (third term) by Constraints (3.11c). We exclude j on the left-hand side to ensure that the constraint is valid. Concerning the later, we can also restrict ourselves to time steps $r \geq t - 2$ since any fire spreading from j to i would necessarily have occurred by $t - 1$. These two constraints model the paths the fire takes from the vertices in F. However, it does not yet ensure correctness as it permits spontaneous ignition of edges that act like a fire breach and allow the fire to be contained without using a single firefighter. For an example, consider the path shown in Figure 1 with $F = \{1\}$ and assume $d = 0$.

By (3.11b), the fire spreads at $t = 1$ along the edge $[1, 2]$, i.e., $x_{12}^1 = 1$. Considering now the edge $[2, 3]$, i.e., $i = 2$ and $j = 3$, and $t = 2$, the left-hand side (lhs) of (3.11c) equals 1 as the fire reached vertex 2 at time $t - 1 = 1$. Thus, we would either need to defend vertex 3 at time 2 or the fire spreads to it along the edge. However, at the third term on the right, if we simply set $x_{43}^2 = 1$ then the right-hand side (rhs) equals 1 and (3.11c) is fulfilled for $i = 2$ and $j = 3$ even with $x_{23}^2 = 0$, cutting off the fire path. Moreover, (3.11c) is also satisfied for all other edges and time periods, especially for the reverse of edge $[2, 3]$, i.e., $i = 3$ and $j = 2$, at $t = 3$ as $x_{43}^2 = 1 = lhs$ and $x_{12}^1 = 1 = rhs$. This allows us to save vertices 4 and 5 without using a single firefighter. To remedy this, we adapted the no-spontaneous-ignition constraints (3.3) for this formulation, resulting in Constraints (3.11d) and (3.11e). Finally, defense and burning are binary by Condition (3.11f) and (3.11g), respectively. Integrality

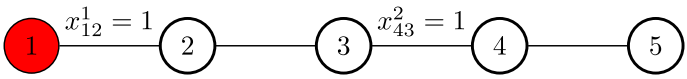


Fig. 1 A path with five vertices

of one of those variables can again be relaxed (follows analogous to Lemma 3.2 using the equivalence $b_i^t = \sum_{r=1}^{t} \sum_{(k,i)\in E} x_{ki}^t$).

Inequalities to strengthen the formulation

As for CF, we can replace (3.1f) by (3.2). Adapting (3.1e), inequalities

$$d_i^t + \sum_{r=1}^{t} \sum_{k\in N(i)} x_{ki}^r \leq 1, \qquad i \in V,\ 1 \leq t \leq T, \tag{3.12}$$

rule out that a node is defended and burning at the same time. Note that they only cut off dominated solutions, see Lemma A.1. Those inequalities can again be formulated just for T

$$d_i^T + \sum_{t=1}^{T} \sum_{k\in N(i)} x_{ki}^t \leq 1, \qquad i \in V, \tag{3.13}$$

or the first $\lceil \alpha\, T \rceil$ periods, denoted as "α(3.12)".

Another proposal for an inequality is to ensure that a fire can not spread in both directions along an edge $[i, j]$ and also not if at least one of the two end nodes is defended:

$$\beta d_i^t + (1-\beta) d_j^t + \sum_{r=1}^{t} x_{ij}^r + \sum_{r=1}^{t} x_{ji}^r \leq 1, \qquad [i, j] \in E,\ 1 \leq t \leq T, \tag{3.14}$$

where $\beta \in [0, 1]$ (for the computational tests, we used $\beta = 0.5$). Again, they only cut off dominated solutions, see Lemma A.2. As before, this inequality can be formulated just for the last period

$$\beta d_i^T + (1-\beta) d_j^T + \sum_{t=1}^{T} x_{ij}^t + \sum_{t=1}^{T} x_{ji}^t \leq 1, \qquad [i, j] \in E, \tag{3.15}$$

or the first $\lceil \alpha\, T \rceil$ periods, denoted as "α(3.14)".

Preprocessing

Moreover, we can make use of several preprocessing steps. Starting with $t = 1$, firstly no fire can spread to an initially burning vertex, so we can set

$$x_{ki}^1 = 0 \qquad i \in F,\ k \in N(i)\,.$$

Secondly, no fire can spread out of a node $i \in V \setminus F$ and no fire can spread into i unless it comes from an initially burning vertex, which allows for every $i \in V \setminus F$

$$x_{ij}^1 = 0, \quad j \in N(i) \qquad \text{and} \qquad x_{ki}^1 = 0, \quad k \in N(i) \setminus F.$$

For $t \geq 2$, firstly a fire cannot spread out of or into vertices in F, so we can set

$$x_{ki}^t = x_{ik}^t = 0, \qquad i \in F,\ k \in N(i),\ 2 \leq t \leq T.$$

Secondly, a fire cannot spread from a vertex i at time t if its distance $dist(i, F)$ to the closest fire is greater than or equal to t, which allows

$$x_{ij}^t = 0, \qquad i \in V,\ j \in N(i),\ 2 \leq t \leq dist(i, F)\,.$$

And thirdly, akin to the previous, a fire cannot spread into a vertex at time t if its distance to the closest fire is strictly greater than t, yielding

$$x_{ki}^{t} = 0, \qquad i \in V,\ k \in N(i),\ 2 \leq t < dist(i, F).$$

3.5 Computational experiments

We coded all integer programming formulations in C++ using CPLEX 20.1 and ran the experiments on a Dell PowerEdge R740 server running Scientific Linux 7 using a single thread with a time limit of one hour.

3.5.1 Data generation

For the number n of nodes, we considered values of 100, 150, 200, 250, and 300. The number of starting fires f and firefighters d for graphs up to 150 nodes ranges from 1 to 3 in steps of 1 and for larger graphs from 1 to 5 in steps of 2. An instance for the problem is fully characterized by the tuple (G, F, d), where G is the underlying graph, F the set of starting fires, and d the number of firefighters.

Starting with the graphs, we drew coordinates for the vertices uniformly at random over the square $[0, 10]^2$. We then computed the Gabriel graph (Ruben Gabriel & Sokal, 1969) and the Delaunay triangulation (Delaunay, 1934) for the point sets. Both graphs are connected and planar, with the Gabriel graph being a proper subgraph of the Delaunay graph. We generated five random graphs for each number of nodes. The average node degree over all generated instances for the Gabriel and Delaunay graphs is 3.5 and 5.3, respectively. For each graph, we then picked the set of starting fires. To avoid trivial cases where a fire starts, e.g., on a leaf or on a vertex with a very high node degree, we sorted the vertices be non-decreasing deviations from the average node degree. That is, the first vertex in the list is the one whose node degree is closest to the average node degree for the graph. In case of ties, vertices with a smaller index come first. Afterwards, we pick the first f vertices in the list to make up the set of starting fires F. Finally, for each pair (G, F) we then set the number of firefighters. Using this hierarchical procedure, the instances $(G, F, *)$ all have identical graphs and starting fires and only differ in the number of firefighters. This allows us to assess how an increase in the number of firefighters affects the computational performance of the considered formulations. Moreover, for the same graph G a larger set of starting fires F includes all smaller sets F', i.e., $F' \subset F$ for $|F'| < |F|$. For each configuration, we generated five instances using different random seeds.

3.5.2 Analysis of the classical formulation CF

We start by analyzing the computational performance of the considered formulations in terms of the underlying graph and the number of starting fires and firefighters for instances with up to 200 nodes. For that we use the original IP formulation for CFG presented in Develin and Hartke (2007) with upper bound T_0^{max}. The results for both types of graphs are shown in Table 1. Note that each reported result is the average computed over five independently generated instances, each created with a distinct random seed. The first column specifies the number of nodes and the second and third columns the number of starting fires and firefighters, respectively. The next five columns present the average run time, the average MIP gap returned by CPLEX, the total number of optimally solved instances for Gabriel

Table 1 Results for the original formulation CF

Nodes	Fires	firefighters	Gabriel Time	Gap	Opt	T^{opt}	#Burn	Delaunay Time	Gap	Opt	T^{opt}	#Burn
100	1	1	16.1	0.00	5	4.8	12.8	191.5	0.00	5	8.4	54.2
		2	2.2	0.00	5	2.2	2.8	13.6	0.00	5	4.0	10.6
		3	1.1	0.00	5	1.4	1.6	4.4	0.00	5	2.2	3.2
	2	1	34.5	0.00	5	6.4	42.2	129.9	0.00	5	6.8	67.2
		2	11.5	0.00	5	4.4	18.2	36.6	0.00	5	4.6	32.2
		3	6.3	0.00	5	2.8	8.2	26.6	0.00	5	4.0	17.8
	3	1	38.8	0.00	5	6.4	51.8	120.4	0.00	5	6.0	74.2
		2	17.0	0.00	5	4.4	28.6	52.3	0.00	5	5.2	51.2
		3	7.4	0.00	5	3.4	17.6	39.1	0.00	5	4.6	34.6
150	1	1	164.9	0.00	5	7.0	33.6	2535.4	13.78	3	11.2	97.0
		2	20.2	0.00	5	3.0	4.4	102.5	0.00	5	4.4	12.0
		3	8.7	0.00	5	2.0	2.0	22.2	0.00	5	2.4	4.2
	2	1	167.1	0.00	5	8.2	56.8	1856.6	0.00	5	8.8	115.4
		2	44.1	0.00	5	4.6	17.8	405.1	0.00	5	6.4	61.0
		3	26.8	0.00	5	3.0	10.0	170.8	0.00	5	5.0	29.0
	3	1	245.7	0.00	5	8.2	81.2	1142.4	0.00	5	7.2	121.6
		2	45.8	0.00	5	5.4	41.8	329.2	0.00	5	6.2	82.8
		3	53.7	0.00	5	4.6	26.2	285.4	0.00	5	5.6	57.0
200	1	1	629.2	0.00	5	6.4	32.0	3471.6	42.08	1	11.6	174.8
		3	14.4	0.00	5	1.6	2.0	103.4	0.00	5	3.2	6.0
		5	2.8	0.00	5	1.0	1.0	13.1	0.00	5	1.4	1.4
	3	1	2369.7	1.29	4	10.0	109.6	3309.9	6.37	1	8.4	167.4
		3	145.0	0.00	5	4.2	23.2	1293.0	0.00	5	6.6	84.2
		5	79.3	0.00	5	1.6	10.8	436.5	0.00	5	1.4	40.4
	5	1	931.7	0.00	5	8.2	136.8	3054.7	2.52	2	8.0	173.8
		3	222.3	0.00	5	5.0	58.8	1311.0	0.00	5	6.2	113.2
		5	132.4	0.00	5	3.6	32.8	543.7	0.00	5	5.0	70.6

graphs, the average game duration, and the average number of burning vertices, respectively. Observe that for the instances where optimality was not proven within the time limit, a solution time equal to the time limit was assigned to compute the average solution time, i.e., the actual solution times are underestimated. The last five columns report the same for Delaunay graphs.

Beginning with the Gabriel instances, a first striking observation is that for fixed n and f, the run times generally decrease monotonically with an increasing number of firefighters, often quite drastically. This is surprising, as–in principle–more firefighters increase the combinatorial complexity of the problem. However, the average game duration and the number

of burning vertices decreases, sometimes quite drastically, with an increasing number of firefighters, as illustrated in columns T^{opt} and "#Burn". Indeed, with more firefighters the fire can be contained more easily and sooner, rendering the formulations easier to solve overall. Fixing only n, the same cannot be said for the number of fires. While $f = 1$ are the easiest instances to solve, of the tested values $f = 3$ are the hardest to solve on average and, e.g., harder than $f = 5$ for $n = 200$. Not unsurprisingly, the run times increase with an increasing value of n. Turning to the Delaunay instances, the first thing to notice is that they are much harder to solve than the Gabriel instances, which is not surprising due to the larger number of edges. Moreover, while the results show identical behavior with respect to an increasing number of firefighters, the computational difficulty of instances changes with respect to the number of fires. The most computationally challenging instances are now for just one starting fire (and one firefighter). Summing up, for both types of graphs the most computationally demanding instances are those for $d = 1$, whereas there is no clear picture for f.

Next, we analyze the impact of the different upper bounds T_l^{max}, $l \in \{0, \ldots, 4\}$. The results for CF, averaged over all numbers of starting fires and firefighters, are shown in Table 2. Note that each reported result is the average computed over 45 instances. The first three columns specify the upper bound T_l^{max} used, the number of nodes, and the average value of T, respectively. The remaining columns are the same as in Table 1. Additionally, for each upper bound, we also report the average run time and gap over all instances with the same number of nodes and the total number of optimally solved instances. As reported in Table 2, compared to T_3^{max} using T_4^{max} allows an overall improvement of approximately 40.28% on the bound of the game duration (from 83.9 to 50.1 time steps on average). For the Gabriel instances, there is a notable drop in the solution time of the formulation from T_1^{max} to T_2^{max} (the latter being the best bound from the literature), while T_2^{max} and T_3^{max} yield similar results. However, this is dwarfed by the improvement from T_2^{max}/T_3^{max} to T_4^{max}, the new bound that we proposed which solves the problem approximately four times faster (the average running time decreases from 170.6 seconds with $T_{\max}^3$ to 49.1 seconds with $T_{\max}^4$). For the Delaunay graphs, we observe a similar behaviour with the improvement of using T_4^{max} again standing out (the average running time decreases from 612.3 seconds with $T_{\max}^3$ to 330 seconds with $T_{\max}^4$), demonstrating a significant computational advantage.

As a result, for the remaining tests we will always use T_4^{max} and we consider instances with different numbers of starting fires, but with only one firefighter, i.e., $d = 1$. Moreover, we are going to look at slightly larger graphs ranging from 150 up to 250 vertices.

First, we will test the effectiveness of the different potential improvements listed in Section 3.2, starting with relaxing the integrality of either b_i^t or d_i^t (§3.2.3). The results, aggregated over all number of nodes, are shown in Table 3. The first two columns specify whether b_i^t or d_i^t are kept binary (Yes) or not (No). The remaining columns are as above. We highlight in bold the overall best run times, gaps, and numbers of instances solved to optimality (we follow the same convention in subsequent tables). Relaxing the integrality of the defense variables d_i^t increases the efficiency of the formulation across all three statistics, in contrast to doing the same for the fire variables b_i^t. Therefore, all remaining tests will be conducted by replacing $d_i^t \in \{0, 1\}$ with $d_i^t \geq 0$.

Finally, we check whether preprocessing (§3.2.1) and any of the changes to the set of constraints are beneficial (§3.2.2): replacing (3.1f) by (3.2), adding (3.3), and removing (3.1e) or replacing it by (3.4) or α(3.1e). For the latter, we chose $\alpha = 10\%$ because T^{opt} was on average around 8-9% of T_4^{max}. We do not report the result of (3.3), as adding the inequality is detrimental across all instances and statistics. For these tests, we have yet again increased the problem size for Gabriel graphs with node numbers ranging from 200 to 300

Table 2 Computational results for different values of T averaged over all f and d

	Nodes	T	Gabriel Time	Gap	Opt	Delaunay Time	Gap	Opt
T_0^{max}	100	100	15.0	0.00	45	68.3	0.00	45
	150	150	86.3	0.00	45	761.1	1.53	43
	200	200	503.0	0.21	44	1504.1	5.66	34
	Total	150	201.4	0.07	134	777.8	2.40	122
T_1^{max}	100	98	13.7	0.00	45	65.8	0.00	45
	150	148	86.8	0.00	45	731.4	1.58	43
	200	197	482.0	0.24	44	1457.8	6.16	34
	Total	147.7	194.2	0.08	134	751.7	2.58	122
T_2^{max}	100	61.3	11.0	0.00	45	53.4	0.00	45
	150	91.7	66.7	0.00	45	624.5	1.44	43
	200	102.3	435.6	0.21	44	1094.2	5.19	35
	Total	85.1	171.1	0.07	134	590.7	2.21	123
T_3^{max}	100	60.1	10.3	0.00	45	54.1	0.00	45
	150	90.7	69.1	0.00	45	665.1	1.63	43
	200	100.9	432.4	0.24	44	1117.6	6.16	34
	Total	83.9	170.6	0.08	134	612.3	2.60	122
T_4^{max}	100	35.8	3.7	0.00	45	17.9	0.00	45
	150	53.8	20.4	0.00	45	259.9	0.49	44
	200	60.7	123.1	0.00	45	712.0	1.15	40
	Total	50.1	49.1	0.00	135	330.0	0.55	129

Table 3 Comparison of relaxing the integrality of b_i^t or d_i^t for $n = 150, 200, 250$

Binary b_i^t	d_i^t	Gabriel Time	Gap	Opt	Delauney Time	Gap	Opt
Yes	Yes	743.8	0.38	42	2046.1	8.18	27
Yes	No	**637.3**	**0.15**	**43**	**1822.1**	**7.43**	**29**
No	Yes	800.1	0.20	42	2130.0	8.14	26

now (the instance size stays the same for Delaunay graphs). The results are shown in Table 4, where the first three columns list the respective changes. Note that each reported result is the average computed over 45 instances for each graph type.

While the impact of preprocessing is marginal, replacing (3.1f) by (3.2) yields better results on all accounts. Concerning (3.1e), replacing the constraints by α(3.1e) is generally helpful, except when using preprocessing and (3.2). While most combinations outperform CF (whose results are in the second row), none of them is best across all graph types and statistics. To put these results into perspective, we include in the final row the results of the best formulation in the literature, namely the one in Ramos et al. (2020), which uses preprocessing and replaces

Table 4 Comparison of different combinations of improvements for CF with $n = 200, 250, 300$ for Gabriel graphs and $n = 150, 200, 250$ for Delaunay instances

Preprocessing	Defense	Changes	Gabriel Time	Gap	Opt	Delaunay Time	Gap	Opt
No	(3.1f)	−(3.1e)	1512.2	1.78	36	1800.0	6.84	32
			1592.3	1.17	37	1822.1	7.43	29
		−(3.1e)+(3.4)	1490.9	1.06	37	1856.1	7.79	29
		−(3.1e)+α(3.1e)	1405.4	1.23	38	1765.3	7.22	30
	(3.2)	−(3.1e)	1252.9	1.74	38	1652.7	4.45	32
			1363.5	1.12	39	1623.1	5.32	34
		−(3.1e)+(3.4)	1463.7	4.49	37	1751.1	5.45	31
		−(3.1e)+α(3.1e)	1230.1	**0.51**	**41**	1610.4	5.43	34
Yes	(3.1f)	−(3.1e)	1531.5	1.74	36	1804.6	6.89	33
			1586.2	0.85	39	1831.2	7.48	29
		−(3.1e)+(3.4)	1499.0	1.06	38	1863.2	7.84	29
		−(3.1e)+α(3.1e)	1458.7	1.11	36	1767.6	7.21	30
	(3.2)	−(3.1e)	1303.4	1.80	39	1721.7	4.43	31
			1224.5	1.00	39	**1595.0**	5.29	**35**
		−(3.1e)+(3.4)	1444.5	3.13	38	1725.5	5.45	32
		−(3.1e)+α(3.1e)	1435.2	1.11	38	1619.1	**4.43**	34
Ramos et al. Ramos et al. (2020)			2632.2	29.29	18	2951.8	16.38	18

(3.1f) by (3.2), but keeps the defense variables binary and employs T_2^{max}. With our tightened formulation and newly proposed bounds, we can solve twice as many instances to optimality in half of the time compared to the formulation in Ramos et al. (2020) that appears on the bottom of the table, highlighting the benefits of our improvements.

3.5.3 Analysis of the two alternative formulations CF_1 and CF_2

Next, we turn to the two alternative formulations, starting with CF_1 (§3.4.1). For both, we again relax the integrality of the defense variables (we tested relaxing either the fire or the defense variables, or neither). Moreover, we again use $n = 200, 250, 300$ ($n = 150, 200, 250$) for Gabriel (Delaunay) graphs, $d = 1$, and $f = 1, 2, 3$ for $n = 150$ and $f = 1, 3, 5$ for larger instances. The results, aggregated over all numbers of nodes, are shown in Table 5. Note that each reported result is the average computed over 45 instances. The first column specifies whether we use preprocessing or not, the second indicates which constraints we use for bounding the number of firefighters, and the third column shows which of the inequalities (3.6), (3.7), (3.8), (3.9), or (3.10) are added. Again, we also consider the case where (3.7) and (3.8) are only added for the first $\alpha\%$ time steps.

Looking at the first two rows, it becomes immediately clear that preprocessing the fire variables significantly reduces the average run times and gaps and increases the number of solved instances. After observing little impact of preprocessing for CF, this improvement is surprising. One possible explanation is that formulations without preprocessing have signifi-

Table 5 Comparison of different combinations of improvements for CF_1 with $n = 200, 250, 300$ for Gabriel graphs and $n = 150, 200, 250$ for Delaunay instances

Preprocessing	Defense	Valid	Gabriel Time	Gabriel Gap	Gabriel Opt	Delaunay Time	Delaunay Gap	Delaunay Opt
No	(3.5b)		1413.0	1.15	32	2683.2	9.26	19
Yes	(3.5b)		363.1	**0.00**	**45**	845.5	2.63	39
		(3.7)	366.3	**0.00**	**45**	864.0	2.19	39
		α(3.7)	314.0	**0.00**	**45**	779.2	1.96	39
		(3.8)	387.5	**0.00**	**45**	953.0	3.11	39
		α(3.8)	**258.8**	**0.00**	**45**	827.9	1.75	**41**
		(3.9)	334.0	**0.00**	**45**	797.5	2.73	40
		(3.10)	346.5	**0.00**	**45**	859.1	2.14	39
	(3.6)		308.5	**0.00**	**45**	907.3	2.32	39
		(3.7)	275.2	**0.00**	**45**	**776.7**	1.74	**41**
		α(3.7)	322.7	**0.00**	**45**	873.6	2.07	40
		(3.8)	394.4	**0.00**	**45**	997.7	2.80	39
		α(3.8)	307.7	**0.00**	**45**	905.7	1.52	39
		(3.9)	335.2	0.19	44	799.3	**1.34**	**41**
		(3.10)	361.5	**0.00**	**45**	883.9	2.96	39

cantly more binary variables and constraints after CPLEX applied its presolve than those with preprocessing. We have carried out a preliminary study to analyse this claim and we obtained that for Gabriel graphs, instances where we apply preprocessing have on average around 7% fewer binary variables and 6.5% fewer constraints after CPLEX presolve than those without preprocessing. For Delaunay instances, the reductions are around 5% and 4.5%, respectively. Therefore, all remaining results for CF_1 are reported with preprocessing. A second observation is that, as a whole, replacing (3.5b) by (3.6) has very little effect. Concerning (3.7) and (3.8), usually the final-period constraints (3.9) and (3.10) or the α% versions yield better results, but not always. However, comparing the results with those for CF in Table 4, CF_1 clearly outperforms CF.

Finally, we consider CF_2 (see §3.4.2). The results, aggregated over all numbers of nodes, are shown in Table 6. Note that each reported result is the average computed over 45 instances. The first column indicates whether we use preprocessing, the second column specifies which constraints we use for bounding the number of firefighters, and the third column shows the combinations of valid inequalities used. We omit (3.12), (3.14), and α(3.14), as these always yielded inferior results compared to either not having them at all or using (3.13) and (3.15).

Starting with preprocessing, while this is generally beneficial for Gabriel instances, the results for Delaunay graphs are mixed. In contrast, using aggregate defense constraints (3.2) improves the results in almost all cases and for both types of graphs. Concerning the inequalities (3.13) and (3.15), for Gabriel graphs including either of the two is helpful, but not both, whereas the latter option is usually best for Delaunay instances. Finally, while CF_2 clearly outperforms CF it falls behind CF_1.

In summary, formulation CF_1 provides the best overall results across all instances and significantly improves on the best-known formulation in the literature. Recall that for Gabriel graphs, the best combination of CF_1 results in a substantial decrease in computation time, from 2632.2 seconds to just 258.8 seconds (a 90.2% reduction), and solves more than twice

Table 6 Comparison of different combinations of improvements for CF_2 with $n = 200, 250, 300$ for Gabriel graphs and $n = 150, 200, 250$ for Delaunay instances

Preprocessing	Defense	Valid	Gabriel Time	Gap	Opt	Delaunay Time	Gap	Opt
No	(3.1f)		1055.3	2.08	37	1716.1	11.11	29
		(3.13)	696.7	0.12	42	1484.1	8.07	31
		(3.15)	713.5	0.53	41	1577.9	6.88	33
		(3.13)&(3.15)	853.2	0.64	41	1402.3	4.64	35
		α(3.12)	691.5	0.34	42	1493.3	7.56	31
	(3.2)		851.6	0.99	39	1704.2	9.32	30
		(3.13)	537.4	0.14	44	1455.8	6.94	30
		(3.15)	572.9	0.20	42	1402.8	5.72	33
		(3.13)&(3.15)	606.9	0.23	43	1214.6	4.59	35
		α(3.12)	530.9	0.33	42	**1121.4**	**4.34**	**36**
Yes	(3.1f)		1020.2	4.52	37	1641.4	9.06	29
		(3.13)	638.5	0.21	43	1491.0	7.61	32
		(3.15)	611.2	0.39	42	1671.4	7.64	31
		(3.13)&(3.15)	745.8	0.14	44	1521.5	6.87	32
		α(3.12)	689.4	0.24	42	1446.0	5.84	31
	(3.2)		842.1	1.32	39	1682.5	9.44	29
		(3.13)	**489.6**	**0.00**	**45**	1382.9	6.59	34
		(3.15)	567.8	0.34	43	1401.4	7.80	33
		(3.13)&(3.15)	661.7	0.12	42	1298.2	6.12	34
		α(3.12)	556.9	0.19	43	1139.8	5.08	**36**

as many instances to optimality (45 vs. 18). This improvement is even more significant considering that the state-of-the-art computational times are underestimated, as explained at the beginning of the section. Similarly, for Delaunay graphs, the best combination of CF_1 reduces the solution time from 2951.8 seconds to 776.7 seconds (a 73.7% reduction), and increases the number of solved instances from 18 to 41.

3.5.4 Analysis of denser graphs

Both Gabriel and Delaunay graphs are planar and, thus, have a bounded average node degree of 6. In order to study the performance of the three formulations for denser graphs, we amended the Delaunay instances as follows. For each Delaunay graph, we randomly added edges to the graph until the average node degree reached a certain value. We used values of 10, 15, and 20 for the desired average node degree (recall that our Delaunay graphs have an average node degree of 5.3). We chose the probability for adding an edge ~~to be~~ inversely proportional to its length. We carried out experiments for CF, CF_1, and CF_2. As there was no single configuration performing best across all types of graphs and formulations, we decided to use the initial formulations with relaxed defense variables, $T_4^{\max}$, and only the aggregate defense constraints, i.e., $CF - (3.1e) + (3.2)$, $CF_1 - (3.5b) + (3.6)$, and $CF_2 - (3.1e) + (3.2)$, but varying the use of preprocessing. The results can be found in

Table 7 below. Each reported result is an average over 15 instances. Specifically, we use $d = 1$, $f = 1, 2, 3$ for $n = 100, 150$, $f = 1, 3, 5$ for $n = 200$, and five distinct random seeds. The first column indicates the average node degree, where the value 5.3 represents the original Delaunay instances. The second and third columns indicate whether the preprocessing is used and the number of nodes, respectively. The fourth and fifth columns show the average game duration and the average percentage of burning vertices, respectively. The next three columns show for CF the average runtime and gap, and the number of optimally solved instances. The next two sets of three columns contain the same information, but this time for CF_1 and CF_2.

We can make several interesting observations. Starting with CF, instances with higher edge densities are computationally more demanding to solve than the Delaunay instances. Since the number of constraints in CF increases with the number of edges, this behavior is intuitive. However, the results for the average game duration—and especially for the average percentage of burning vertices—are surprising. Starting from an average node degree of 10, almost all vertices catch fire, which makes identifying optimal solutions essentially trivial. One possible explanation is that the instances are highly degenerated with many optimal solutions, creating problems for the MIP solver. A second interesting observation pertains to the use of preprocessing for CF_1. We have seen that preprocessing is very beneficial for this formulation, but for Delaunay instances to a lesser extent than for Gabriel graphs. The current results confirm this trend, with preprocessing having less and less impact for an increasing edge density. A final observation relates to CF_2, which has shown a better performance than CF but a worse performance than CF_1 for Gabriel and Delaunay instances. For denser graphs, this completely turns around with CF_2 considerably outperforming CF and CF_1 for average node degrees of 15 and 20. This is very surprising, as increasing the number of edges not only increases the number of constraints but also the number of binary variables in CF_2. A possible explanation is that CF_2 is more efficient at dealing with degenerate instances where almost every node is burning.

4 Cost-value firefighter

In this section, we present the cost-value firefighter game (CVFG). To that end, we denote the cost of defending vertex $i \in V$ by $c_i \in \mathbb{R}^+$ and its value for being saved by $v_i \in \mathbb{R}^+$. The defense budget for each time step is denoted by $B \in \mathbb{R}^+$ and we assume that $c_i \leq B$ for all $i \in V$ and $c_i = v_i = 0$ for all $i \in F$. CVFG adheres to the same rules as CFG, except for replacing the second rule by:

2. In every time step $t > 0$, we can spend any amount between 0 and B on improving the defense of a node, as long as the combined spending per time step across all vertices does not exceed B.
   The defense of a vertex can be built up gradually, i.e., over multiple time steps. A vertex counts as defended iff the total amount of defense budget spent on the vertex so far is equal to or exceeds its defense cost c_i.

The goal of the game is to spend the available defense budget across vertices in each time step so as to maximize the value of non-burning vertices at the end of the game. We point out that this variant allows us to model the case in which the fire takes more than one period to spread between two adjacent nodes $i, j \in V$, $(i, j) \in E$. To achieve this, we insert as many auxiliary nodes into (i, j) as the number of time steps required for the contagion to spread,

Table 7 Computational results for denser graphs

Degree	Pre	Nodes	T^{opt}	Burn	$CF-(3.1e)+(3.2)$			$CF_1-(3.5b)+(3.6)$			$CF_2-(3.1e)+(3.2)$		
					Time	Gap	Opt	Time	Gap	Opt	Time	Gap	Opt
5.3	No	100	7.1	65%	30.5	0.00	15	43.5	0.00	15	52.5	0.00	15
		150	9.2	70%	378.7	0.00	15	542.1	0.00	15	555.5	1.38	14
		200	9.5	76%	1522.9	1.79	13	1633.5	4.99	11	1551.8	4.96	11
				Total	644.0	0.60	43	739.7	1.66	41	719.9	2.11	40
	Yes	100	7.1	65%	29.4	0.00	15	17.8	0.00	15	45.7	0.00	15
		150	9.2	70%	374.2	0.00	15	166.0	0.00	15	551.2	1.20	14
		200	9.5	76%	1517.8	1.87	13	874.9	0.91	14	1478.4	3.88	12
				Total	640.5	0.62	43	352.9	0.30	44	691.7	1.69	41
10	No	100	3.5	96%	132.5	0.00	15	78.2	0.00	15	64.8	0.00	15
		150	4.0	97%	1335.5	0.12	12	1128.4	0.02	13	992.5	0.03	13
		200	3.8	98%	2752.2	0.83	8	2100.9	0.19	11	1308.3	0.04	13
				Total	1406.7	0.32	35	1102.5	0.07	39	788.6	0.02	41
	Yes	100	3.5	96%	122.3	0.00	15	60.7	0.00	15	57.6	0.00	15
		150	4.0	97%	1378.5	0.15	12	872.2	0.01	14	966.5	0.03	12
		200	3.8	98%	2750.7	0.81	8	1516.4	0.05	13	1432.5	0.04	12
				Total	1417.1	0.32	35	816.4	0.02	42	818.9	0.02	39

Table 7 continued

Degree	Pre	Nodes	T^{opt}	Burn	$CF-(3.1e)+(3.2)$			$CF_1-(3.5b)+(3.6)$			$CF_2-(3.1e)+(3.2)$		
					Time	Gap	Opt	Time	Gap	Opt	Time	Gap	Opt
15	No	100	2.9	97%	103.5	0.00	15	43.8	0.00	15	17.7	0.00	15
		150	3.1	98%	1038.3	0.00	15	799.3	0.00	15	105.1	0.00	15
		200	3.3	98%	2996.5	0.63	5	2101.6	0.02	12	780.1	0.00	15
				Total	1379.4	0.21	35	981.6	0.01	42	301.0	0.00	45
	Yes	100	2.9	97%	104.9	0.00	15	40.0	0.00	15	18.4	0.00	15
		150	3.1	98%	1130.5	0.00	15	687.9	0.00	15	105.1	0.00	15
		200	3.3	98%	2922.8	0.54	6	1524.5	0.01	14	701.9	0.00	15
				Total	1386.1	0.18	36	750.8	0.00	44	275.1	0.00	45
20	No	100	2.5	98%	56.3	0.00	15	31.0	0.00	15	24.2	0.00	15
		150	2.9	98%	913.8	0.00	15	564.5	0.00	15	81.8	0.00	15
		200	2.7	99%	2126.8	0.27	10	1720.2	0.07	11	265.5	0.00	15
				Total	1032.3	0.09	40	771.9	0.02	41	123.8	0.00	45
	Yes	100	2.5	98%	49.8	0.00	15	32.0	0.00	15	25.5	0.00	15
		150	2.9	98%	907.7	0.00	15	571.8	0.00	15	83.8	0.00	15
		200	2.7	99%	2291.2	0.28	10	1755.3	0.12	10	269.8	0.00	15
				Total	1082.9	0.09	40	786.4	0.04	40	126.4	0.00	45

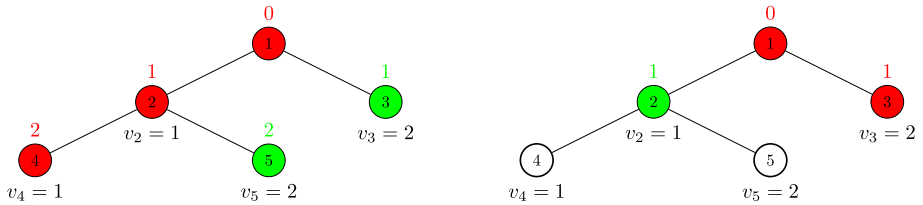


Fig. 2 Example of different game durations for optimal defense strategies. Vertices in red (green) have caught fire (been defended) where the red (green) number next to it indicates the time they did so

minus one. For such auxiliary nodes, we set a profit equal to zero and a defense cost equal to the maximum of the defense costs of i and j (or, for simplicity, to ∞).

For an optimal defense strategy, we again denote by T^{opt} the time step when the game ends. If there is more than one optimal defense strategy, then we pick the one with the smallest T^{opt}. An example of multiple optimal defense strategies with differing T^{opt} is shown in Figure 2. Let $B = 1$, $c_i = 1$, $i \in V$, and $f = 1$ with the initial fire starting in vertex 1. The numbers in red or green above the nodes show the time step when a node catches fire or has been defended, respectively. The left-hand side depicts an optimal defense strategy with duration 2, while the right-hand side shows an optimal defense strategy with duration 1. In both strategies, the value of non-burning vertices is 4.
Analogously to CFG, we can prove that:

Lemma 4.1 *For an instance of Cost-Value Firefighter, there exists an optimal defense strategy S such that the maximum defense budget is used every time step up to $T^{opt} - 1$ and only on vertices that would not be saved without being defended in any time step up to T^{opt}.*

Proof The proof is analogous to the one for Lemma 4.1 after replacing the number of (un)used firefighters by the (un)used defense budget. □

4.1 Integer programming formulation

In the following, we present the first mixed-integer linear programming formulation for the Cost-Value Firefighter game which is based on CF, see Section 3. Let $T \in \mathbb{Z}^+$ be an upper bound for T^{opt}. In addition to the defense and fire variables d_i^t and b_i^t, respectively, we define the decision variables

$$s_i^t \;=\; \text{the total amount of defense budget spent on vertex } i \text{ up to and including time } t,$$

for all $i \in V$ and $1 \le t \le T$. With these variables, we can formulate the problem as

$$(CVF) \quad \min \sum_{i \in V} v_i \, b_i^T \tag{4.1a}$$

$$\text{s.t.} \quad b_i^{t-1} \le b_i^t, \qquad i \in V,\ 1 \le t \le T, \tag{4.1b}$$

$$d_i^{t-1} \le d_i^t, \qquad i \in V,\ 2 \le t \le T, \tag{4.1c}$$

$$b_i^{t-1} \le b_j^t + d_j^t, \qquad i \in V,\ j \in N(i),\ 1 \le t \le T, \tag{4.1d}$$

$$\sum_{i \in V} \left(s_i^t - s_i^{t-1} \right) \le B, \qquad 1 \le t \le T, \tag{4.1e}$$

$$c_i \cdot d_i^t \le s_i^t, \qquad i \in V,\ 1 \le t \le T, \tag{4.1f}$$

$$s_i^0 = 0, \qquad i \in V, \tag{4.1g}$$
$$b_i^0 = 1, \qquad i \in F, \tag{4.1h}$$
$$b_i^t \in \{0, 1\}, \qquad i \in V,\ 0 \le t \le T, \tag{4.1i}$$
$$d_i^t \in \{0, 1\}, \qquad i \in V,\ 1 \le t \le T, \tag{4.1j}$$
$$0 \le s_i^t \le c_i, \qquad i \in V,\ 0 \le t \le T. \tag{4.1k}$$

The objective function (4.1a) minimizes the value of burning vertices. Constraints (4.1b)−(4.1d) are identical to (3.1b)−(3.1d) for CF. Constraints (4.1e) ensure that the firefighters adhere to the defense budget of B per time step. Conditions (4.1f) and (4.1g) enforce that a sufficient defense budget is spent on a vertex to defend it and that no vertices start with any defense budget spent towards them, respectively. Constraints (4.1h) and (4.1i) are identical to (3.1g) and (3.1i) and, finally, defense is integer by (4.1j) and the defense budget spent on a vertex must be non-negative by (4.1k).

4.2 Improvements to the formulation

Analogous to CFG, we can again replace the period-by-period budget constraint (4.1e) by an aggregated version

$$\sum_{i \in V} s_i^t \le t \cdot B, \qquad 1 \le t \le T, \tag{4.2}$$

and, thus, also drop (4.1g). The equivalence of the two sets of constraints is straightforward: Any optimal solution for CVF trivially satisfies (4.2). Vice versa, if for an optimal solution S of CVF−(4.1e)−(4.1g)+(4.2) we have $\sum_{i \in V}(s_i^t - s_i^{t-1}) > B$ for some t, then we can simply reassign the excess $\sum_{i \in V}(s_i^t - s_i^{t-1}) - B$ in t to a previous period (which must be possible as (4.2) holds for all $t = 1, \ldots, T$) and no vertex will be defended later than in S.

Next, we can again add inequalities

$$b_i^t + d_i^t \le 1, \qquad i \in V,\ 1 \le t \le T, \tag{4.3}$$

for every time period, or just for the last

$$b_i^T + d_i^T \le 1, \qquad i \in V, \tag{4.4}$$

or the $\alpha\%$ first periods, labelled as α(4.3). By the same reasoning as in §3.2.2 for dropping (3.1e), these only cut off dominated solutions. Moreover, we can add the inequalities

$$s_i^{t-1} \le s_i^t, \qquad i \in V,\ 1 \le t \le T, \tag{4.5}$$

which ensure that the defense of a node is monotone increasing. Adding these inequalities to CVF will only cut off dominated solutions and they can even replace (4.1c):

Lemma 4.2 *Adding constraints* (4.5) *to CVF will only cut off dominated solutions and they can even replace* (4.1c) *without losing correctness.*

Proof Starting with the first statement (adding constraints (4.5) to CVF will only cut off dominated solutions), assume that we have an optimal solution to CVF that ends the game after T^{opt} such that $s_i^{t-1} > s_i^t$ for at least one $i \in V$ and $1 \le t \le T^{opt}$, i.e., some of the defense budget spent previously on a node $i \in V$ was redirected in period t to some other node(s). But then we must have $d_i^t = d_i^{t-1} = 0$ due to constraints (4.1c), (4.1f), and (4.1k). Hence, instead of spending the amount $s_i^{t-1} - s_i^t$ on defending i before time t, it could have

already been redirected to those other nodes which needed it in t. Thus, we can find an alternative optimal defense strategy where $s_i^{t-1} \leq s_i^t$.

Concerning the second statement (constraints (4.5) can replace (4.1c)), for an optimal solution to formulation CVF−(4.1c)+(4.5) we can only have $d_i^t = 1$ if $s_i^t = c_i$, as $s_i^t \leq 1$ by (4.1k). Hence, once the latter holds, $d_i^r = 1$ remains feasible for $r = t+1, \ldots, T$ due to (4.5) and we can therefore construct an optimal solution that satisfies (4.1c). □

Finally, as for CF, we can preprocess the fire variables and relax the integrality of b_i^t or d_i^t without losing the correctness of the formulation.

Lemma 4.3 *If we remove the binary conditions on the fire (defense) variables in CVF, then we can convert any optimal solution to the relaxed formulation into one in which all fire (defense) variables have binary values.*

Proof The proof follows analogously to the one for Lemma 3.2 after replacing d by B and 1 by c_i. □

Concerning inequalities to strengthen the formulation, according to Lemma 4.1 we do not need to spend anything on a vertex that will not be defended in the end

$$s_i^T = c_i \cdot d_i^T, \qquad i \in V. \tag{4.6}$$

To develop the next family of inequalities, we consider a permutation γ that sorts the vertices in order of non-decreasing defense costs, i.e., $c_{\gamma(1)} \leq c_{\gamma(2)} \leq \ldots \leq c_{\gamma(n)}$. Note that the first f costs are 0, as they belong to the vertices initially burning. For $1 \leq t \leq T$, let $k^t \leq n - f$ be the largest value for which the following inequality holds:

$$\sum_{r=1}^{k^t} c_{\gamma(r+f)} \leq t \cdot B.$$

Then k^t is the maximum number of nodes that can possibly be defended up to time step t. Therefore, we can include the following family of inequalities, which are in fact valid for CVF by the above reasoning:

$$\sum_{i \in V} d_i^t \leq k^t, \quad 1 \leq t \leq T\,. \tag{4.7}$$

4.3 Bounds on the maximum duration of the game

While the first two bounds $T_0^{max} = n$ and $T_1^{max} = n - f$ for CFG trivially carry over, we will refer to them as $T_{c,0}^{\max}$ and $T_{c,1}^{\max}$ hereafter to prevent ambiguity. The next three do not carry over because they are based on the number of firefighters. In the following, we derive four new upper bounds for T^{opt}. The first two are adaptations of $T_3^{max} = \lceil \frac{n-f}{d} \rceil$ and $T_4^{max} = \lceil \frac{n-f}{d+1} \rceil$ for CFG, namely $T_{c,2}^{\max} = \lceil \sum_{i \in V} c_i / B \rceil$ and $T_{c,3}^{\max} = \lceil \sum_{i \in V} c_i / (B + \min_{i \in V \setminus F} c_i) \rceil$. The former computes how many time steps are needed to defend all vertices. The latter additionally takes into account that at least one new vertex must ignite every time step for the game to continue.

The next bound is a slightly different adaptation of T_4^{max} for CFG to CVFG. For that, we consider a permutation σ of $\{1, \ldots, n\}$, such that $c_{\sigma(1)} \geq c_{\sigma(2)} \geq \ldots \geq c_{\sigma(n)}$. Let r be the largest integer such that $\sum_{k=1}^{r} c_{\sigma(k)} \leq B$, i.e., r is the minimum number of vertices that can be defended per time step (as $c_i \leq B$, we have $r \geq 1$). Then, we obtain

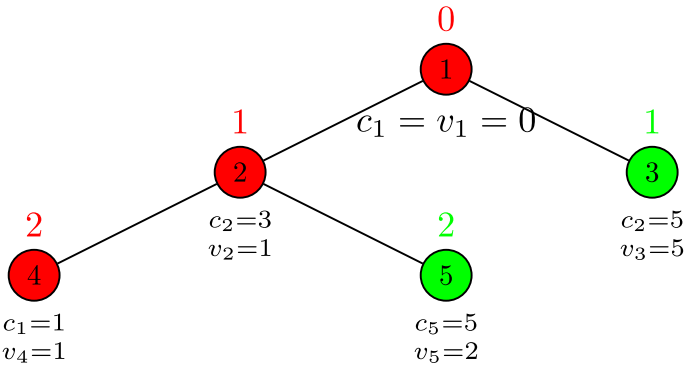


Fig. 3 Graph for Example 4.5

Lemma 4.4 *For the cost-value firefighter game, we have* $T^{opt} \leq T^{max}_{c,4} = \lceil \frac{n-f}{r+1} \rceil$ *where*

$$r = \max \left\{ l \mid \sum_{k=1}^{l} c_{\sigma(k)} \leq B \right\}.$$

Proof Assume there exists an instance where $T^{opt} > \lceil \frac{n-f}{r+1} \rceil$. To start with, we observe that after $\lceil \frac{n-f}{r+1} \rceil$ time steps, at least $r \cdot \lceil \frac{n-f}{r+1} \rceil$ vertices must have been defended. Otherwise, we could have defended the newly burning or defended vertex at time T^{opt} already before. This leaves at most $(n-f) - r\lceil \frac{n-f}{r+1} \rceil$ vertices that can still burn, and this number is bounded by

$$(n-f) - r\left\lceil \frac{n-f}{r+1} \right\rceil \leq (n-f) - r\frac{n-f}{r+1} = \frac{n-f}{r+1} \leq \left\lceil \frac{n-f}{r+1} \right\rceil.$$

However, as $T^{opt} > \lceil \frac{n-f}{r+1} \rceil$ and at least one new vertex has to burn every time step up to and including $T^{opt} - 1$, by the end of time $T^{opt} - 1$ all vertices are either burning or have been defended, which contradicts T^{opt} being the minimum. □

The following example illustrates the bounds that we have derived so far.

Example 4.5 Consider the graph shown in Figure 3, with $f = 1$, $B = 5$, and the defense costs and node values as shown. The fire starts at vertex 1, and the red and green numbers show an optimal defense strategy. At time $t = 1$, all budget is spent on defending node 3, with a value of 5. Afterwards, vertex 2 catches fire. In the next time step, it is optimal to defend node 5, after which the game ends, i.e. $T^{opt} = 2$.

Concerning our bounds, we have $T^{max}_{c,0} = 5$, $T^{max}_{c,1} = 4$, $T^{max}_{c,2} = \lceil \frac{14}{5} \rceil = \lceil 2.8 \rceil = 3$, and $T^{max}_{c,3} = \lceil \frac{14}{6} \rceil = \lceil 2.\bar{3} \rceil = 3$. While $T^{max}_{c,2} = T^{max}_{c,3}$, we can see the effect of adding the smallest c_i in the ratio before rounding. Finally, as $r = 1$ we have $T^{max}_{c,4} = \lceil (n-f)/(r+1) \rceil = 2$, i.e., the bound is tight.

To derive our final improved upper bound, we again consider the permutation γ introduced in Section 4.2. Let $1 \leq t \leq n - f$. If

$$\sum_{r=t+f+1}^{n} c_{\gamma(r)} \leq t \cdot B,$$

then we have enough budget to defend the $n-(t+f+1)+1 = n-(t+f)$ most expensive nodes within t periods. Finding the smallest t for which this holds gives us the following bound:

Lemma 4.6 *For the cost-value firefighter game, we have* $T^{opt} \leq T^{max}_{c,5}$ *where*

$$T^{max}_{c,5} \;=\; \min\left\{ t \;\middle|\; \sum_{r=t+f+1}^{n} c_{\gamma(r)} \leq t \cdot B \right\}.$$

Proof By definition

$$\sum_{r=T^{\max}_{c,5}+f+1}^{n} c_{\gamma(r)} \leq T^{\max}_{c,5} \cdot B.$$

Since at least one new node must ignite at every time step $1, \ldots, T^{\max}_{c,5}$ in S, resulting in at least $T^{\max}_{c,5}$ newly burning vertices by the end of period $T^{\max}_{c,5}$. Taking into account the f initially burning vertices, the game has to end at time $T^{\max}_{c,5}$ (at the latest), as we have by definition sufficient budget to defend the remaining $n-(T^{\max}_{c,5}+f)$ non-burning nodes of strategy S by the end of $T^{\max}_{c,5}$. Then at the time $T^{\max}_{c,5}$ the game has finished. □

Example 4.5 (cont'd) *For* $t = 1$, *we have*

$$\sum_{r=t+f+1}^{n} c_{\gamma(r)} \;=\; \sum_{r=3}^{5} c_{\gamma(r)} \;=\; 3+5+5 \;=\; 13 \;\not\leq\; 5 \;=\; t \cdot B\,.$$

However, for $t = 2$ *we have* $c_{\gamma(4)} + c_{\gamma(5)} = 10 \leq 10 = 2 \cdot 5$. *Hence,* $T^{max}_{c,5} = 2$, *which is again tight.*

While we obviously have $T^{\max}_{c,1} \leq T^{\max}_{c,0}$ and $T^{\max}_{c,3} \leq T^{\max}_{c,2}$, the remaining relationships are unclear. We will numerically compare all these bounds and assess their computational impact in Section 4.5.

4.4 Alternative formulations for CVFG

Our first alternative formulation for CVFG mirrors CF_1 for CFG (see Section 3.4.1).

$$(\mathrm{CVF}_1) \qquad \min \quad \sum_{i\in V}\sum_{t=0}^{T} v_i\, \hat{b}^t_i \tag{4.8a}$$

$$\text{s.t.} \quad \text{(3.5c)}, \text{(3.5d)}, \text{(3.5e)}, \text{(3.5f)}, \text{(4.1e)}, \text{(4.1g)}, \text{(4.1k)},$$

$$c_i \cdot \sum_{r=1}^{t} \hat{d}^r_i \leq s^t_i, \qquad i \in V,\ 1 \leq t \leq T, \tag{4.8b}$$

where the last constraint is the adaptation of (4.1f). Likewise, the second alternative formulation for CVFG mirrors CF_2.

$$(\mathrm{CVF}_2) \quad \min \quad \sum_{i\in V}\sum_{j\in N(i)}\sum_{t=1}^{T} v_j\, x^t_{ij} \tag{4.9a}$$

s.t. (3.1c), (3.11b), (3.11c), (3.11d), (3.11e), (3.11f), (3.11g), (4.1e), (4.1f), (4.1g), (4.1k).

Concerning improvements, preprocessing, and inequalities, everything mentioned in Section 3.4 carries over, except for inequalities related to the number of firefighters. Moreover, we can again replace (4.1e) by (4.2).

4.5 Computational experiments

We again coded the mixed-integer programming formulations in C++ using CPLEX 20.1 and ran the experiments on a Dell PowerEdge R740 server running Scientific Linux 7 using a single thread with a time limit of one hour. Moreover, the graphs were generated as before. Concerning defense costs c_i and node values v_i, $i \in V$, we drew the former uniformly at random from the interval $[5, 25]$. The latter were then drawn uniformly at random from the interval $[0.5c_i, 1.5c_i]$, i.e., roughly proportional to the defense cost. Finally, the defense budget was set according to $B_d = 25(d + 1)/2$, where d was chosen as in Section 3.5. For each type of graph and each combination of n, f, and d, we generated five random instances.

Analysis of the formulation CVF

We start again by analyzing the computational performance of the formulations in terms of the underlying graph and the number of starting fires and defense budget for instances with up to 200 nodes. The results for both types of graphs and $T = T^{\max}_{c,0}$ are shown in Table 8. The headings are the same as in Table 1, except for "Feas" which denotes the number of instances where CPLEX could find a feasible solution within the time limit. The entries are averaged over five instances each, and the number d determines the budget B_d as defined above.

Comparing the results with those in Table 1, we can see that CVFG is much more computationally challenging to solve than CFG. For some instances with $n = 200$, CPLEX cannot even find a feasible solution. As for CFG, however, for a fixed n and f the run times generally decrease with an increasing defense budget, often quite drastically. Fixing only n, yet again the same cannot be said for the number of fires, although there is a tendency of $f = 3$ yielding the most computationally challenging instances. Finally, analogously to CFG, the Delaunay instances are much harder to solve than the Gabriel instances.

Next, we compare the different upper bounds $T^{\max}_{c,l}$, $l \in \{1, \ldots, 5\}$. In Table 9 we present the values for $n = 200$ and the Gabriel instances. The entries are averaged over five instances. The results for $n = 100$ and $n = 150$ as well as for Delaunay instances are very similar. The first two columns denote the number of fires and defense budget, and the remaining ones show the average values of the different upper bounds (each is over five instances). There is a notable drop from $T^{\max}_{c,1}$ to $T^{\max}_{c,2}$, while the others are much closer together. $T^{\max}_{c,5}$ has the lowest averages and also provided the smallest value of T for every single instance. $T^{\max}_{c,3}$ is slightly worse on average, but still matched the best T in seven out of 45 instances. Recall that reducing the bound on the maximum duration of the game has a great impact on the formulation, as each unit reduction implies fewer variables and constraints.

Finally, in Table 10 we present the results for solving CVF with $T^{\max}_{c,2}$, the best trivial bound for CVFG (and the straightforward adaptation of the one in Ramos et al. (2020)), and for $T^{\max}_{c,5}$, the most sophisticated of our new bounds, but only for B_1. The entries are averaged over five instances. We can observe a significant reduction in average run times already for $T^{\max}_{c,2}$ compared with $T^{\max}_{c,0}$, but this is dwarfed by the results for $T^{\max}_{c,5}$. Not only can CPLEX

Table 8 Detailed results for formulation CVF with $T = T_{c,0}^{\max}$

Nodes	Fires	d for B_d	Gabriel Time	Gap	Feas	Opt	Delaunay Time	Gap	Feas	Opt
100	1	1	26.8	0.00	5	5	304.3	0.00	5	5
		2	20.2	0.00	5	5	322.8	0.00	5	5
		3	3.9	0.00	5	5	81.3	0.00	5	5
	2	1	95.5	0.00	5	5	325.4	0.00	5	5
		2	50.0	0.00	5	5	168.6	0.00	5	5
		3	37.0	0.00	5	5	141.2	0.00	5	5
	3	1	123.2	0.00	5	5	297.3	0.00	5	5
		2	100.0	0.00	5	5	189.9	0.00	5	5
		3	60.6	0.00	5	5	116.0	0.00	5	5
150	1	1	484.1	0.00	5	5	2070.9	33.93	5	3
		2	78.7	0.00	5	5	2148.1	38.24	5	3
		3	10.1	0.00	5	5	700.5	0.00	5	5
	2	1	1153.2	0.00	5	5	1987.2	15.80	5	4
		2	769.3	0.00	5	5	1680.0	17.76	5	4
		3	514.2	0.00	5	5	2161.6	16.26	5	4
	3	1	1109.4	0.00	5	5	1904.8	16.42	5	4
		2	934.3	0.00	5	5	1482.5	17.25	5	4
		3	1247.2	0.00	5	5	1659.5	15.98	5	4
200	1	1	2854.2	38.55	5	3	3600.1	94.33	3	0
		2	202.9	0.00	5	5	1936.3	39.19	5	3
		3	4.2	0.00	5	5	119.6	0.00	5	5
	3	1	3600.1	90.22	2	0	3600.3	37.36	4	0
		2	3600.2	34.60	3	0	3321.8	18.34	5	2
		3	3369.6	39.91	5	1	3034.1	53.87	5	2
	5	1	3378.2	13.28	5	2	3600.1	25.08	5	0
		2	2920.4	7.34	5	4	3320.9	14.58	5	2
		3	2407.6	0.00	5	5	3091.2	0.24	5	4

now find a feasible solution for every instance, it can even optimally solve all but one within one hour.

As a result, for the remaining tests we will always use $T_{c,5}^{\max}$ and we consider instances with different values for the number of starting fires and B_1. Moreover, we are going to look at larger graphs ranging from 150 up to 250 vertices.

Next, we will test the effectiveness of relaxing the integrality of b_i^t or d_i^t. The results, aggregated over all number of nodes and starting fires, are shown in Table 11. The entries are averaged over 45 instances and again we highlight in bold the overall best run times, gaps, and

Table 9 Comparison of the upper bounds $T^{\max}_{c,k}$ with $n = 200$

Fires	d for B_d	Gabriel $T^{\max}_{c,1}$	$T^{\max}_{c,2}$	$T^{\max}_{c,3}$	$T^{\max}_{c,4}$	$T^{\max}_{c,5}$
1	1	199.0	119.8	100.0	100.0	87.4
	3	199.0	60.2	54.8	67.0	52.6
	5	199.0	40.4	37.6	50.0	37.0
3	1	197.0	117.4	97.8	99.0	86.0
	3	197.0	59.4	54.2	66.0	51.8
	5	197.0	39.6	37.4	50.0	36.6
5	1	195.0	117.8	98.0	98.0	86.0
	3	195.0	59.2	53.8	65.0	51.4
	5	195.0	39.6	37.0	49.0	36.6
Average		197.0	72.6	63.4	71.6	58.4

numbers of instances solved to optimality. The columns are as above. Slightly surprisingly, relaxing the defense variables is no longer the best option. While there is no clear winner, overall, it seems best to keep both variables binary. The tests for CVF_1 and CVF_2 yielded similar results, so we will also keep both variables binary for those formulations.

Finally, we check whether preprocessing or any of the changes to the set of constraints is beneficial: replacing (4.1e) by (4.2), adding (4.3), α(4.3), or (4.4), adding (4.5) or replacing (4.1c) by it, or adding (4.6) or (4.7). We begin by testing each change individually. The results are shown in Table 12, where the first column denotes the respective Change. The entries are averaged over 45 instances.

We report the results for CVF in the first row (change: none) and then for the suggested changes. While most changes provide some improvement, they are usually lesser and not across all graph types or statistics.

However, this is only half of the story. At the end of the table, we report the results for combining some of the changes (we tested (almost) all combinations, but only present the most promising ones as the full table would take too much space, and many combinations are clearly worse than the selection we present here). It is striking that combining several changes, which may each be minor in themselves, results in huge improvements. All combinations have in common that they replace the initial with the aggregate budget constraints and add some version of (4.3) (although there is no clear indication of which one works best overall), and in two cases also (4.6).

In summary, the proposed procedure to tighten the formulation leads to a substantial improvement, with average run times decreasing from 1189.6 seconds to 339.7 seconds, representing a 71.44% reduction for Gabriel graphs and from 1874.7 seconds to 902 seconds, representing a 51.89% reduction for Delaunay graphs. This highlights the significant computational efficiency gained by tightening the formulation.

4.5.1 Analysis of the two alternative formulations CVF_1 and CVF_2

Next, we turn to the two alternative formulations, starting with CVF_1. We adapted (4.6) and (4.7) straight-forwardly to CVF_1. The results, aggregated over all numbers of nodes, are shown in Table 13.

Table 10 Results for formulation CVF with B_1

		$T_{c,2}^{\max}$						$T_{c,5}^{\max}$					
Nodes	Fires	Gabriel Time	Gap	Opt	Delaunay Time	Gap	Opt	Gabriel Time	Gap	Opt	Delaunay Time	Gap	Opt
100	1	6.4	0.00	5	56.4	0.00	5	2.3	0.00	5	29.7	0.00	5
	2	54.5	0.00	5	145.7	0.00	5	15.9	0.00	5	67.3	0.00	5
	3	52.0	0.00	5	181.9	0.00	5	22.8	0.00	5	94.9	0.00	5
150	1	47.0	0.00	5	219.6	0.00	5	33.2	0.00	5	185.9	0.00	5
	2	637.9	0.00	5	810.9	0.00	5	75.8	0.00	5	268.7	0.00	5
	3	431.6	0.00	5	514.1	0.00	5	135.8	0.00	5	278.1	0.00	5
200	1	416.9	0.00	5	3600.0	39.45	0	245.3	0.00	5	2061.0	0.00	5
	3	1919.6	1.35	4	3165.2	0.35	4	648.1	0.00	5	1385.0	0.00	5
	5	1934.0	0.00	5	3235.8	1.69	1	907.1	0.00	5	2497.7	0.06	4

Table 11 Comparison of relaxing integrality of b_i^t or d_i^t with $n = 150, 200, 250$

Binary		Gabriel				Delaunay			
b_i^t	d_i^t	Time	Gap	Feas	Opt	Time	Gap	Feas	Opt
Yes	Yes	1100.7	**3.52**	45	**38**	1760.4	**6.37**	45	**33**
Yes	No	**1003.1**	3.99	45	37	1751.7	11.31	43	32
No	Yes	1046.8	4.01	45	**38**	**1700.6**	9.17	45	**33**

Table 12 Comparison of different combinations of improvements for CVF with $n = 150, 200, 250$

Change	Gabriel Time	Gap	Opt	Delaunay Time	Gap	Opt
none	1189.6	7.56	36	1874.7	13.08	30
preprocessing	1175.8	8.12	37	1928.2	14.38	29
− (4.1e) + (4.2)	1096.6	7.39	37	1898.0	13.08	31
+ (4.3)	1019.1	3.92	36	1686.3	10.39	31
+ (4.4)	1033.6	6.10	37	1723.4	10.60	32
+ α(4.3)	1030.7	6.34	38	1879.2	13.08	31
+ (4.5)	988.7	4.54	38	1861.7	12.56	30
− (4.1c) + (4.5)	1111.4	4.70	38	1936.2	16.21	30
+ (4.6)	1112.6	6.74	37	1860.7	12.15	31
+ (4.7)	1132.6	6.06	37	1836.0	8.43	31
−(4.1e)+(4.2)+(4.3)	438.8	0.44	44	1021.0	3.78	39
−(4.1e)+(4.2)+α(4.3)	356.6	0.01	44	1072.0	1.19	39
−(4.1e)+(4.2)+α(4.3)+(4.6)	477.9	0.04	44	1006.3	1.45	**42**
preproc−(4.1e)+(4.2)+(4.3)+(4.6)	454.3	0.44	44	**902.0**	2.21	41
preproc−(4.1e)+(4.2)+(4.4)	473.5	0.17	44	1056.8	3.00	41
preproc−(4.1e)+(4.2)+α(4.3)	**339.7**	**0.00**	**45**	1021.2	**0.68**	**42**

Table 13 Comparison of different combinations of improvements for CVF_1 with $n = 150, 200, 250$

Change	Gabriel Time	Gap	Opt	Delaunay Time	Gap	Opt
None	922.7	1.10	37	1950.0	4.35	26
preprocessing	397.7	0.03	44	1131.6	0.94	37
preproc−(4.1e)+(4.2)	201.6	**0.00**	**45**	427.7	0.04	43
preproc−(4.1e)+(4.2)+(4.4)	176.3	**0.00**	**45**	404.2	0.03	**44**
preproc−(4.1e)+(4.2)+(4.3)+(4.6)	171.5	**0.00**	**45**	535.4	0.19	42
preproc−(4.1e)+(4.2)+(3.7)	192.5	**0.00**	**45**	**395.2**	0.05	43
preproc−(4.1e)+(4.2)+α(3.7)+(4.7)	**153.5**	0.03	44	411.3	0.23	43
preproc−(4.1e)+(4.2)+(4.4)+(4.6)+(4.7)	210.4	**0.00**	**45**	496.5	**0.01**	**44**

Looking at the first two rows, it becomes immediately clear that preprocessing the fire variables again significantly reduces the average run times and gaps and increases the number of solved instances. at Row 3, a further significant improvement can be achieved by replacing (4.1e) with (4.2). All further changes help to improve the results in one way or another, as shown in the last five rows of the table, but none of them stands out as being best across all graph types and statistics. Summing up, the best combinations for CVF_1 can halve the runtimes of those for CVF and increase the number of optimally solved Delaunay instances from 42 to 44 (in both cases, the unsolved instance is `Delauney_250_4_3_1`, which is the fourth instance with 250 nodes, 3 starting, fires, and B_1).

Finally, we consider CVF_2. The results, aggregated over all numbers of nodes, are shown in Table 14 (which has the same structure and columns as Table 13). Then, the entries are averaged over 45 instances.

While preprocessing is not beneficial, using the aggregate defense constraints (4.2) improves the results for Gabriel graphs, but not so much for Delaunay instances. The best results are again obtained by combining various changes. Finally, while CVF_2 outperforms CVF, it falls again behind CVF_1. Therefore, CVF_1 provides the best overall results across all instances.

In summary, we show that refining the bound on the maximum duration of the game significantly reduces the computation time. Specifically, our bound reduces the computation time by half compared to the direct adaptation of the previously published bound for this new problem. This improvement highlights the efficiency gained through our approach. Furthermore, we propose new formulations and additional inequalities to strengthen the model. We achieve substantial computational benefits when comparing our approach to the direct adaptation of the existing formulation of CFG for CVFG using the best-developed bound on the maximum duration of the game. Specifically, our approach leads to an 87.1% reduction in computation time for the Gabriel graph and a 78.91% reduction for the Delaunay graph. These results emphasize the effectiveness of our refinements in improving computational performance.

4.5.2 Analysis of denser graphs

To compare the formulations for an increasing number of edges, we followed the same procedure as in Section 3.5.4 to generate denser graphs and we carried out experiments for CVF, CVF_1, and CVF_2. As there was no single configuration performing best across all types of graphs and formulations, we decided to use the initial formulations with binary fire and defense variables, $T^{max}_{c,5}$, the aggregate defense constraints, and valid inequalities (4.4), i.e., for $CVF/CVF_1/CVF_2$ − (4.1e) + (4.2) + (4.4), and only varying the use of preprocessing. The results can be found in Table 15 below, which has exactly the same structure as Table 7. Each reported result is an average over 15 instances; specifically, we use $d = 1$ for B_d, $f = 1, 2, 3$ for $n = 100, 150$, $f = 1, 3, 5$ for $n = 200$, and five distinct random seeds.

We can make several interesting observations. The first mirrors the one in Section 3.5.4. All three formulations are computationally more demanding to solve for instances with higher edge densities than for the Delaunay instances, despite the fact that most vertices catch fire. A second interesting observation pertains to the use of preprocessing for CVF_1. Unlike CF_1, preprocessing is also very beneficial for higher edge densities, although to a lesser extent. A final observation relates to CVF_2, which again outperforms CVF and CVF_1.

Table 14 Comparison of different combinations of improvements for CVF_2 with $n = 150, 200, 250$

Change	Gabriel Time	Gap	Opt	Delaunay Time	Gap	Opt
None	586.5	2.09	40	1235.3	1.64	37
preprocessing	668.6	2.01	37	1414.4	2.78	33
−(4.1e)+(4.2)	493.8	0.56	42	1185.6	1.86	36
+(4.5)	761.8	2.34	39	1491.7	2.63	32
−(4.1c)+(4.5)	596.3	0.36	42	1133.9	2.09	37
−(4.1e)+(4.2)−(4.1c)+(4.5)+(4.4)	285.8	0.09	44	661.7	0.52	41
−(4.1e)+(4.2)−(4.1c)+(4.5)+(4.4)+(4.6)+(4.7)	296.0	**0.00**	**45**	719.1	0.39	41
preproc−(4.1e)+(4.2)−(4.1c)+(4.5)+(4.4)+(4.6)+(4.7)	319.4	0.03	44	695.8	0.34	**43**
preproc−(4.1e)+(4.2)−(4.1c)+(4.5)+(4.4)	259.9	**0.00**	**45**	863.6	0.41	40
preproc−(4.1e)+(4.2)−(4.1c)+(4.5)+(4.4)+(4.7)	**229.0**	**0.00**	**45**	**658.3**	**0.25**	**43**

Table 15 Computational results for denser graphs with formulations $CVF/CVF_1/CVF_2 - (4.1e) + (4.2) + (4.4)$

Degree	Pre	Nodes	T^{opt}	Burn	CVF Time	Gap	Opt	CVF_1 Time	Gap	Opt	CVF_2 Time	Gap	Opt
5.3	No	100	5.53	45%	23.9	0.00	15	72.0	0.00	15	39.3	0.00	15
		150	5.27	26%	86.7	0.00	15	168.8	0.00	15	84.7	0.00	15
		200	7.20	49%	721.2	0.00	15	1463.8	0.10	14	717.7	0.02	14
				Total	277.3	0.00	45	568.2	0.03	44	280.6	0.01	44
	Yes	100	5.53	45%	23.8	0.00	15	26.2	0.00	15	41.7	0.00	15
		150	5.27	26%	85.7	0.00	15	64.8	0.00	15	94.5	0.00	15
		200	7.20	49%	692.8	0.00	15	300.6	0.00	15	758.5	0.00	15
				Total	267.4	0.00	45	130.5	0.00	45	298.3	0.00	45
10	No	100	3.60	93%	117.5	0.00	15	70.1	0.00	15	66.0	0.00	15
		150	4.13	94%	1907.3	0.87	11	1192.6	0.00	15	979.9	0.00	15
		200	4.07	96%	2860.8	2.43	6	2568.4	2.93	7	1487.8	0.81	11
				Total	1628.5	1.10	32	1277.1	0.98	37	844.6	0.27	41
	Yes	100	3.60	93%	102.3	0.00	15	33.3	0.00	15	47.3	0.00	15
		150	4.13	94%	1849.4	0.69	11	564.4	0.00	15	779.3	0.00	15
		200	4.07	96%	2689.6	2.60	6	2217.9	1.11	8	1562.7	0.62	11
				Total	1547.1	1.10	32	938.5	0.37	38	796.4	0.21	41

Table 15 continued

Degree	Pre	Nodes	T^{opt}	Burn	CVF Time	Gap	Opt	CVF_1 Time	Gap	Opt	CVF_2 Time	Gap	Opt
15	No	100	2.93	95%	51.4	0.00	15	47.0	0.00	15	33.0	0.00	15
		150	3.20	96%	1186.0	0.06	14	835.6	0.00	15	425.4	0.00	15
		200	3.13	97%	2346.8	0.72	10	1733.2	1.53	12	1282.0	0.24	13
				Total	1194.7	0.26	39	871.9	0.51	42	580.1	0.08	43
	Yes	100	2.93	95%	43.9	0.00	15	14.7	0.00	15	34.2	0.00	15
		150	3.20	96%	924.8	0.04	14	399.4	0.00	15	362.1	0.00	15
		200	3.13	97%	2051.4	0.37	11	1410.0	0.66	12	1246.8	0.28	13
				Total	1006.7	0.14	40	608.1	0.22	42	547.7	0.09	43
20	No	100	2.47	95%	27.2	0.00	15	25.6	0.00	15	38.9	0.00	15
		150	2.93	96%	672.8	0.00	15	560.4	0.00	15	230.2	0.00	15
		200	2.73	97%	1623.2	0.40	10	1489.1	0.43	12	863.9	0.05	14
				Total	774.4	0.13	40	691.7	0.14	42	377.7	0.02	44
	Yes	100	2.47	95%	22.6	0.00	15	12.1	0.00	15	29.0	0.00	15
		150	2.93	96%	418.1	0.00	15	166.5	0.00	15	156.4	0.00	15
		200	2.73	97%	1547.5	0.21	10	861.6	0.73	14	835.2	0.00	15
				Total	662.8	0.07	40	346.7	0.24	44	340.2	0.00	45

5 Conclusions

In this paper, we analyzed the classical firefighter game under a mathematical optimization approach. Moreover, we introduced a new variant called the cost-value firefighter game which considers heterogeneous vertex values and defense costs, i.e., the fixed number of firefighters has been replaced by a flexible budget of defense. For both problems, new mixed-integer programming formulations were developed together with new (valid) inequalities and tighter bounds on the duration of the game.

Computational experiments show that our formulations significantly improve the performance of existing approaches in the literature. The proposed (valid) inequalities and bounds increase the number of instances solved to optimality within the time limit and reduce the computing time, making the models much more applicable for larger instances. These advances could further benefit matheuristic approaches that incorporate the problem formulation, potentially improving their efficiency and solution quality. Among the different formulations we proposed, the one that performed best in non-degenerate games for both problems was the approach that used variables that indicate the time period when the status of a node changes.

This work opens several avenues for future research considering alternative problem variants, such as dynamic budget allocation or network disruptions, which could further enrich the model's applicability. Moreover, future research could explore advanced methodological strategies, such as parametric or decomposition-based approaches, to further exploit the advantages of the results presented in this paper.

A Proofs for inequalities

Lemma A.1 *Inequalities* (3.12)

$$d_i^t + \sum_{r=1}^{t} \sum_{k \in N(i)} x_{ki}^r \leq 1, \qquad i \in V,\ 1 \leq t \leq T,$$

only cut off dominated solutions of formulation CVF$_2$.

Proof Assume first that $\sum_{r=1}^{t} \sum_{k \in N(i)} x_{ki}^r \geq 2$ for an $i \in V$ and $1 \leq t \leq T$. Let t' be the first time period for which $x_{ki}^{t'} = 1$ for a $k \in N(i)$, i.e., the period when the fire first reaches i. But then, we can set all other $x_{ki}^r = 0$, as it is enough for the fire to reach i from just one of its neighbours and by (3.11c) the fire has already spread from i to all adjacent $j \in N(i) \setminus \{k\}$ in period $t' + 1$ that are undefended or not yet burning.

Assume now that $d_i^t = 1$ and $x_{ki}^{t'} = 1$ for a $k \in N(i)$ and $1 \leq t' \leq t$. If $t' < t$, then i has already caught fire before time t (unless it was already defended) and, thus, there is not point in defending i at time t. If, instead, $t' = t$, then considering the edge $[k, i]$ in (3.11c), we can set $x_{ki}^t = 0$ as i has been defended in t. □

Lemma A.2 *Inequalities* (3.14)

$$\beta d_i^t + (1 - \beta) d_j^t + \sum_{r=1}^{t} x_{ij}^r + \sum_{r=1}^{t} x_{ji}^r \leq 1, \qquad [i, j] \in E,\ 1 \leq t \leq T,$$

only cut off dominated solutions of formulation CVF$_2$.

Proof Using the same argument as in the proof of Lemma A.1, we can assume that the third and fourth sum are each less than or equal to 1.

Assume now that $\sum_{r=1}^{t} x_{ij}^{r} + \sum_{r=1}^{t} x_{ji}^{r} \geq 2$ for an $[i, j] \in E$ and $1 \leq t \leq T$. Let $1 \leq r, r' \leq t$ with $x_{ij}^{r} = x_{ij}^{r'} = 1$. Without loss of generality, $r \leq r'$. If $r = r'$, for the fire to spread from i and j, it already had to reach both nodes at period $r - 1$. Hence, the third term on the right-hand side of (3.11c) is already 1, so there is no need for setting either variable to 1 at time r. If, instead $r < r'$, then for the fire to spread from i to j at time r, it must have reached i at time $r - 1$, so there is no need in (3.11c) for the fire to spread from j to i at time $r' > r$.

Finally, assume that $d_i^t = 1$ and some $x_{ij}^{r} = 1$ for $r \leq t$. For the fire to spread from i at time r, it must have reached it by time $r - 1$. Hence, there is no point in defending i at time $t \geq r$. The same holds for the other combinations of defense variable and sum of fire-path variables. □

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Declarations

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